

Reviewed Preprint

v1 • December 31, 2025

Not revised

Reviewed Preprint

v2 • March 18, 2026

Revised by authors

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Competing interests:

No competing interests declared

Funding:

See [page 22](#)

Reviewing editor:

Jean-Jacques Orban de Xivry, KU Leuven, Belgium

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How individual vigor shapes human-human physical interaction

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eLife Assessment

This is an **important** study showing that movement vigor is not solely an individual property but emerges through interaction when two people are physically linked. The evidence is **convincing**, supported by a well-controlled experimental design and modeling that closely match the observed behavior. While the authors provided a helpful comparison of several candidate models of human-human interaction dynamics, the statistical power remains limited.

<https://doi.org/10.7554/eLife.109781.2.sa4>

Abstract

The speed of voluntary movements varies systematically, with some individuals moving consistently faster than others across different actions. These variations, conceptualized as vigor, reflect a time-effort-accuracy tradeoff in motor planning. How do two mechanically coupled partners with different individual vigors collaborate, e.g. to move a table together? Here, we show that such dyads coordinate goal-directed movements with minimal interaction force, exhibiting a *dyadic vigor* with similar characteristics as individual vigor. The emerging dyadic motor plan is strongly influenced by the slower partner, whose vigor predicts dyadic vigor, with effects lasting beyond practice. Computational modeling with stochastic optimal control reveals the critical role of partners' movement timing's uncertainty and vigor in shaping coordination, allowing to predict dyadic movements from individual behavior across diverse conditions. These findings shed light on the mechanisms underlying human collaboration, and may be used in applications ranging from physical training and rehabilitation to collaborative robotics for manufacturing.

1 Introduction

When observing people carrying out actions, one may notice that they move at different pace, with some individuals being systematically slow or systematically fast. Recent research has confirmed this observation [1], identifying *vigor*, an idiosyncratic trait across actions that characterizes the tradeoff between the energy expenditure, accuracy, and time to perform a given goal-oriented movement [2–5], reflecting the value individuals associate with it [6]. At the neural level, vigor may be influenced by reward-driven dopamine secretion in the basal ganglia [7–11], where individuals more sensitive to reward tend to move faster. The relative vigor of an individual within a population has been reported to be remarkably stable across repeated measurements compared to its variability between people [12]. Based on this stability, the duration of self-paced human movements can be predicted across tasks and environmental dynamics, as a tradeoff between minimizing a cost of time and the effort required to complete an action with the desired accuracy [2, 3, 13, 14]. This idiosyncratic cost of time can be identified using inverse optimal control techniques from the standard relationship observed between motion amplitude and duration [15]. In this scheme, the *motor plan*, i.e. the time series of control inputs, emerges from the costs that the central nervous system (CNS) internally minimizes.

When people with different levels of vigor perform a task together, how do they combine their individual motor plans to collaborate efficiently? A pioneering study [16] found that *dyads* of connected partners performing point-to-point arm movements tend to move faster than each individual alone. Nevertheless, this result may have been influenced by an explicit reward associated with decreasing movement time given to dyads. Current knowledge suggests that individuals can identify others' vigor through visual observation [1], but it is unclear whether this is possible when mechanically connected with a partner through an object, such as when moving a table together. In fact, it has been suggested that during point-to-point movements with a prescribed duration, mechanically coupled individuals may move independently as they do not have sufficiently rich information to identify the partner's motion plan [17]. Importantly, even if they could coordinate their motor plans when removing the time prescription, the remaining question is whether they would do this in a systematic way, with some dyads moving consistently faster than others across conditions. This paper investigates how dyads of connected partners perform point-to-point movements without time constraints, how partners coordinate their motor plans, and how their individual vigor shapes the joint movement.

Several mechanisms may drive the movement coordination, or absence thereof, between mechanically coupled individuals. First, each individual may execute their motor plan independently, causing large interaction efforts and quite arbitrary movement strategies for the dyad, a scenario we refer to as the *co-activity* hypothesis [17]. However, one could also expect that, when time is not prescribed, a leader naturally emerges and imposes their preferred motor plan to the dyad. Specifically, a commonly suggested interaction strategy [18] is that the slower (respectively, faster) individual in a dyad could infer and align with the faster (slower) partner's movement strategy [16,19,20], which we refer to as *leader-follower* hypothesis. More recent studies have shown that connected individuals can use interaction forces to exchange their motion plans and both improve performance in a tracking task [21, 22]. These results suggest that connected individuals may co-adapt to establish a common motor plan using haptic communication. A negotiated motor plan could then emerge, where both partners contribute to setting the dyad's behavior, based on a weighted average of their nominal vigor in the task. Hence both partners may similarly contribute to the dyad's strategy by blending their original intents, a hypothesis we refer to as *weighted adaptation*. Alternatively, uncertainty about a partner's motor plan could change an individual's own motor plan and vigor, leading to an *interactive adaptation* hypothesis. According to this view, the dyad's motor plan would not originate from a weighted averaging of the partners' initial strategies, but from a dynamic change of their strategies in response to the current interaction dynamics, which can be represented in cost functions. These four hypotheses, schematized in Fig. 1A [23], yield distinct behavioral predictions that will be tested in the present study.

2 Results

To understand how dyads' partners coordinated their motor plans and test the above hypotheses, we investigated point-to-point movements without timing constraint or explicit reward. Relatively large targets were used in order to relax accuracy constraints. The partners carried out these movements while being connected through a virtual elastic band implemented using individual wrist exoskeletons, as illustrated in Fig. 1B [24]. This setup allowed us to systematically investigate the impact of the interaction on movement planning, by varying movement amplitudes and connections' stiffness, as well as the effort associated with movements using a viscous load. 20 participants (10 dyads) were requested to perform self-paced flexion and extension movements of the right wrist, with amplitudes between 18° and 90° , by controlling a cursor on their individual monitors from A_0 towards one of the targets $\{A_1, A_2, A_3, A_4, A_5\}$ and back to A_0 . The individual cursor position was an affine function of the corresponding wrist angle. The two partners of a dyad first performed movements in a *solo session* so that we could identify their individual motor strategies, in particular their vigor, prior to being connected in a *dyadic session* with an elastic band of stiffness $\kappa = 0.5$ Nm/rad (KL) or $\kappa = 1.6$ Nm/rad (KH). To facilitate reading the figures show

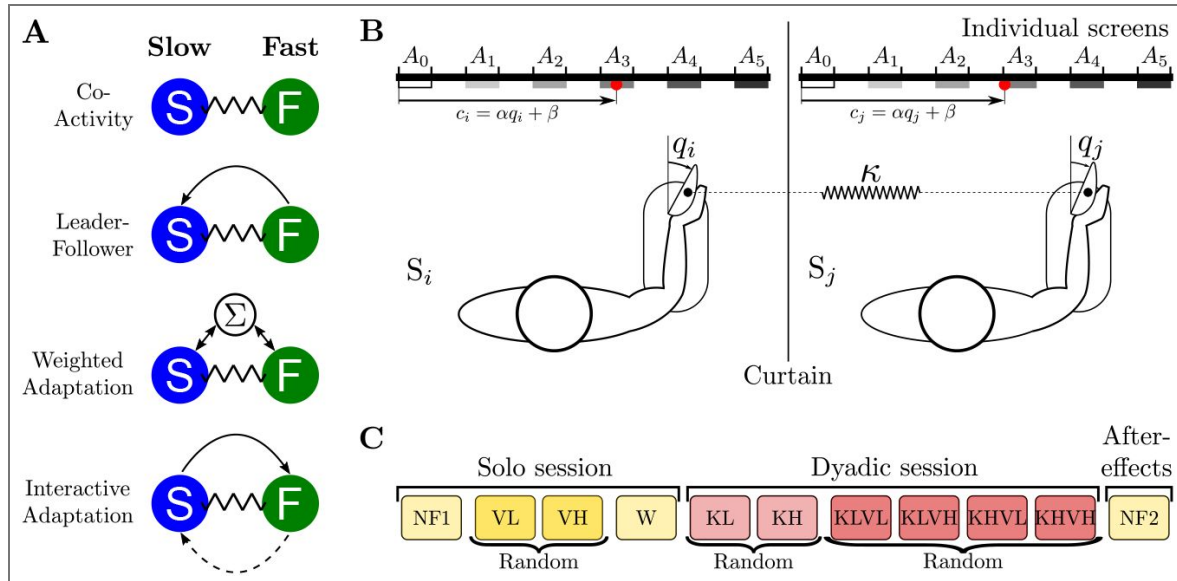


Figure 1. Hypothetical strategies to combine motor plans and experiment.

A. Different hypotheses for coordination between a fast (F) and a slow (S) partner. The arrows represent the haptic communication flow triggering motor adaptations in the respective hypotheses. The *co-activity* strategy corresponding to independent motor plans, *leader-follower* here based on the faster partner's individual motor plan, *weighted adaptation* generalizing the LF hypothesis based on a weighting of their initial strategies in the task, and *interactive adaptation* where both partners dynamically adapt their original motor plan, possibly differently as illustrated by the solid and dashed arrows, due to the interaction with an uncertain partner. **B.** Experiment to investigate vigor in individuals and in mechanically connected dyads. The two partners have to reach one of targets $\{A_1, \dots, A_5\}$ on their individual monitor using wrist flexion/extension of the right arm. Their real-time wrist angles q_i and q_j are mapped to individual red cursors c_i and c_j . In the dyadic session, their hands are coupled through a virtual elastic band of either low (KL) or high (KH) stiffness. **C.** Experimental protocol. The initial solo session consists of four blocks: one null-field block (NF) with exoskeletons' motor off to estimate the individual vigor, two blocks with low (VL) or high (VH) resistive viscous load to vary the cost of movement (i.e. effort), and a null-field washout block (W). The subsequent dyadic session involves six blocks in coupled mode, and a final block in null-field mode to analyze after-effects of the practice with mechanical connection. The KL and KH blocks, performed in random order, first allow participants to familiarize with the interaction, and the KxVx blocks are to investigate all combinations of the connection stiffness and viscous load.

results obtained with the low-stiffness connection KL (except when the two stiffness and viscous loads led to different behaviors), while complementary figures with large connection stiffness KH and other parameters referred to in the text are provided in the Supplementary materials.

2.1 Solo session

We first analyzed participants' behavior in the solo session to verify that vigor, as defined in previous studies [2, 12, 23], is an idiosyncratic trait within our sample, and to identify their motor plan as a baseline for comparison and modeling in the dyadic session. As illustrated in Fig. 2A, the average trajectories in the *null-field condition* (NF1, with motor off) exhibit a bell-shaped velocity profile, where movement duration increases with amplitude. Submovements near the target appear for the farther targets $\{A_4, A_5\}$, as was also observed in self-paced movements of large amplitude carried out with another exoskeleton [3].

Since vigor is expected to be a stable individual trait across effort levels [12, 23], we first verified that VL and VH imposed sufficient resistance to influence movement duration (Friedman: $W = 0.65$, $p < 10^{-3}$, Supplementary Fig. S.1A). Participants moved significantly slower in VL and VH compared to NF1, and slower in VH than in VL ($p < 2 \cdot 10^{-3}$; Cohen's $D > 0.56$ for all of VL-NF1, VH-NF1, VL-VH). These findings confirm that the selected viscous loads required sufficient effort to impact movement duration and analyze the robustness of inter-individual differences.

Next, as shown in Fig. 2B, movement time increased linearly with amplitude for each participant ($r^2 = 0.92 \pm 0.09$) and for the average participant in our sample ($r^2 = 0.99$). We leveraged the extracted affine amplitude-duration relationships to compute a vigor score v_i for each participant relative to the population [3, 23, 24]:

$$v_i = \frac{\sum_{k=1}^5 \bar{T}^2(A_k)}{\sum_{k=1}^5 T_i(A_k) \bar{T}(A_k)}, \quad (1)$$

where $\bar{T}(A_k)$ is the average movement duration of the population for the k^{th} movement amplitude in the relevant null-field condition (NF1 for the solo session, KL or KH for the dyadic session), and $T_i(A_k)$ is the average movement duration of participant i with amplitude A_k in the considered condition. The vigor score is positive, and a participant moving at the same speed as the population average has a unit vigor.

A decomposition of variance revealed that inter-individual variability of the vigor scores through the NF1, VL and VH conditions accounted for 81.2% of the total variance, despite the possible effects of the viscous load on the total variance. This result is consistent with prior experiments testing the timing stability over multiple sessions [12], rather than by varying the effort in the present study. The vigor scores for NF1, VL, VH exhibit distributions similar to previous studies [3, 23] (Supplementary Fig. S.1B) and correlate across conditions (Fig. 2C; $p < 10^{-3}$; $r > 0.71$, Pearson tests), confirming the robustness of the individual vigor score across effort levels.

In summary, the solo session data — characterized by affine amplitude-duration relationships, greater inter-than intra-individual variability, and stable vigor scores across conditions with varied effort levels — confirm that vigor is an idiosyncratic trait in our sample of participants, consistent with prior findings [12, 23].

2.2 Dyadic session

Coordination of motor plans within dyads

To analyze coordination, we identified the slow and fast partner in each dyad based on their vigor in the first null-field block, thereby forming the *slow partners* group and *fast partners* group. We first examined whether reaction times differed between these two groups in the initial null-field block, as it could naturally lead to a leader-follower behavior, with the leader being the participant starting earlier. However, no significant difference was observed ($p > 0.14$, reaction time about 225 ms on average), thus the reaction time did not depend on the group in our task.

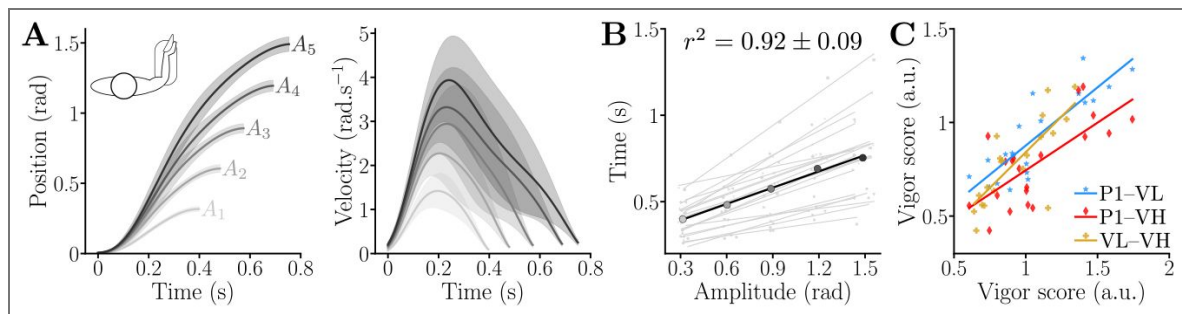


Figure 2. Kinematics and vigor in the solo conditions.

A. Trajectories of reaching movements of different amplitude averaged across the population in the first null-field block (NF1) of Fig. 1C (shaded areas represent the standard deviations). **B.** Resulting individual and averaged amplitude-duration relationships. **C.** The vigor scores in the three solo conditions {NF1,VL,VH} are all correlated, indicating that vigor can be defined robustly across varied effort conditions.

We then assessed whether mechanical coupling affected the participants' reaction time in KL and KH. We found a main effect of the condition on the reaction time for both groups (in both cases: $W > 0.39$, $p < 0.02$). In KL, reaction times were not significantly impacted. In KH, the reaction times of the slow group were significantly shorter than in the initial null-field block ($p = 0.03$, $D = 1.46$), with a similar trend observed for the fast group ($p = 0.06$, $D = 1.03$). However, the fast and slow participants still had similar reaction times ($p > 0.34$ for both KL and KH), with an average of 190 ms.

Regarding trial-by-trial adaptations, the only clear observed trend was an increase in the interaction force during the initial movements, most pronounced in KH and for the largest amplitude in KL (see Supplementary Fig. S.3). This increase in force was not accompanied by changes in movement duration, although slight decreasing trends were occasionally visible across trials (see Supplementary Fig. S.4). These results suggest that participants maintained a minimal level of interaction force to facilitate synchronization.

To test the *co-activity* hypothesis, we modeled the fast and slow partners as planning trajectories independently with their respective preferred movement duration. The individual *co-activity* trajectories and interaction force were computed by assuming participants optimally tracked their desired trajectory with minimum effort and error, while connected via an elastic band of stiffness KL or KH (see Methods) [25]. The averaged partners' trajectory (solid black line in Fig. 3A) exhibits a non-smooth slope change, due to the submovement and stabilization of the fast partner's within the target (solid green curve in Fig. 3A), and produced a relatively large interaction torque that increased with vigor difference between the partners.

The data of connected participants in our experiment did not correspond to these predictions. First, averaged participants' trajectories in the connected session exhibited bell-shaped velocity profiles smoother than in solo performances (Fig. 3B), as analyzed with spectral smoothness metrics [26] (see Supplementary Fig. S.7). Second, the absolute interaction torque was low across all conditions, irrespective of target, viscosity, or connection stiffness (Fig. 3C, Supplementary Figs. S.6C & S.8–S.17). It was slightly increasing with movement amplitude with the low connection stiffness (significant effect of the condition and interaction with the target: $W > 0.32$, $p < 0.013$, confirmed by pairwise comparisons, $p < 0.02$, Cohen's $D > 0.36$ when pooling all KL conditions for each target) but was always less than 0.1 Nm ($p < 0.012$, Cohen's $D > 3.73$ in the six connected conditions), and at maximum close to 0.05 Nm for target A_5 with KL. Interestingly, interaction torques were higher for KL than KH (in all cases: $p < 0.002$, $D > 1.34$), likely due to deteriorated haptic communication with a more compliant connection [27]. Furthermore, interaction torque showed no significant correlation with differences in individual vigor within dyads (KL conditions: $|r| < 0.43$, $p > 0.22$, KH conditions: $|r| < 0.18$, $p > 0.61$; Fig. 3D, Supplementary Fig. S.6D). In contrast to these results, the *co-activity* hypothesis predicted average interaction torques for A_5 of 0.1 Nm with the KL connection (i.e. around twice higher than observed) and 0.23 Nm with KH (i.e. around ten times higher than observed). These findings rule out the *co-activity* hypothesis. They rather suggest that dyadic partners closely coordinated their motor commands, and that at least one of them adapted their individual motor plan.

Characteristics of individual movements within connected dyads

As shown in Fig. 3E (and Supplementary Fig. S.6B), the movement duration of connected participants increased linearly with amplitude, both with low and high connection stiffness, for each participant (KL: $r^2 = 0.85 \pm 0.21$; KH: $r^2 = 0.92 \pm 0.09$) as well as for the average participant (KL: $r^2 = 0.98$; KH: $r^2 = 0.99$). Furthermore, the individual movement duration of connected participants was affected by VL and VH loads ($W > 0.85$, $p < 10^{-3}$ for both KL and KH conditions, as illustrated by Fig. 3F and Supplementary Fig. S.6E). As in solo trials, dyads moved slower when $v > 0$ ($p < 10^{-3}$, Cohen's $D > 1.72$ in all cases) and with VH compared to VL ($p < 10^{-3}$, Cohen's $D > 1.02$ for both KL and KH). Despite potential difficulties in coordinating with the partner, movement duration was not affected by connection stiffness, independently of the required level of effort. Moreover, movement duration was not significantly different between dyads' partners (Fig. 3G)

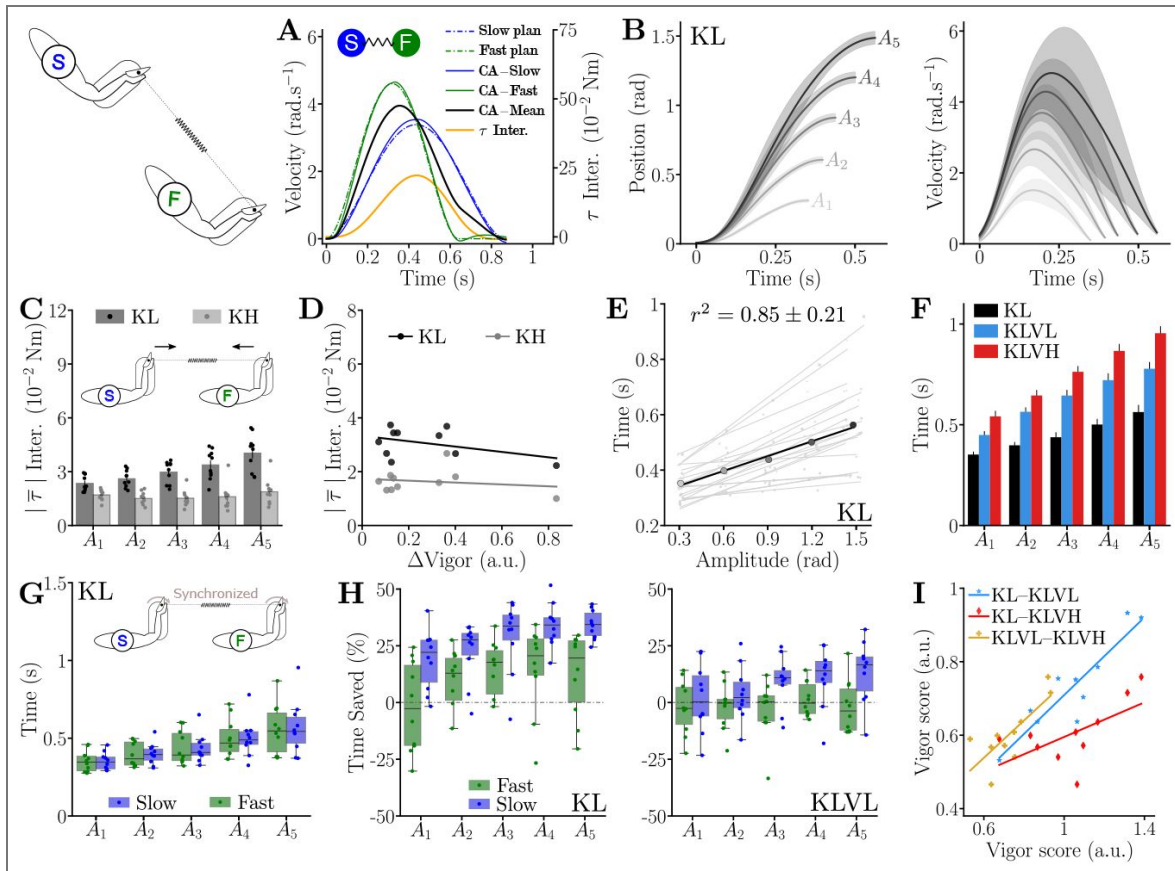


Figure 3. Dyads kinematics, interaction torque and dyadic vigor.

A. Predicted trajectories and interaction torque of a dyad with independent motor plans for the faster and slower partners for A_5 in the KL condition. **B.** Participant's trajectories averaged across the population for KL and for the five targets (shaded areas represent the standard deviations). **C.** Average absolute interaction torque with low (KL) and high (KH) connection stiffness. **D.** The average interaction efforts in KL and KH are independent on the difference in individual vigor between the partners of a dyad (here during the first null-field block). **E.** Amplitude-duration relationships of each participant's movements and average relationship across the population during the KL condition. **F.** Effect of the two different viscous loads on movement duration, for the KL connection, averaged across all participants. **G.** Movement durations are not different between the fast and slow groups in the connected conditions (here KL). **H.** Percentage of time saved by the fast and slow groups between NF1 and KL, and between VL and KLVH, where positive values indicate faster movements. **I.** Correlations between the vigor scores obtained during the three KL connected conditions. Vigor scores were computed using the average movement duration between members of the dyad.

and Supplementary Fig. S.5, no main effect, $p = 0.86$). Altogether, these results indicate that the viscous loads were sufficient to impact the dyads' behavior, and dyads were able to plan and coordinate their movements efficiently in all conditions.

A striking observation is that, when using KL and KH without viscous load, connected participants from both the slow and fast groups moved faster than in the solo session ($W = 0.49$, $p < 10^{-3}$), with a greater change in movement duration for the slow partners (Kruskal $H = 106$, $p < 10^{-3}$ see Fig. 3H [↗](#), Supplementary Fig. S.6F). Post-hoc comparisons confirmed this effect ($p < 10^{-3}$, Cohen's $D > 0.79$ for both groups). With the resistive load, the slow partners still saved time compared to solo trials (Fig. 3H [↗](#), Supplementary Fig. S.6G–I, $W = 0.55$, $p < 10^{-3}$), whereas the fast partners movements durations were not significantly different from the solo conditions (Supplementary Fig. S.6H,I). Post-hoc comparisons confirmed that the slow partners saved time for all the connected conditions ($p < 0.008$, Cohen's $D > 0.71$ in all cases). Importantly, regardless of whether dyads saved time or not, it did not affect their movement smoothness or accuracy (see Supplementary Fig. S.7 and associated analyses).

In summary, dyads moved faster than lone individuals when there was no viscous load. This increased movement speed was a strong marker of unloaded dyadic movements, with, for instance, the slow partners moving 30% faster for movements to target A_5 and the fast partners 20% faster (medians in Fig. 3H [↗](#)). With viscous load, the slow group saved time while the fast group kept their movement duration constant (medians in Fig. 3H [↗](#)). The combination of all these results on interaction efforts and time supports the existence of a shared motor plan in dyad's members, which plan is sometimes clearly differing from the lone motor plan of both participants.

Dyadic vigor

As connected partners coordinated their movements and exhibited affine amplitude-duration relationships, we computed vigor scores based on the average movement duration of the two partners of each dyad using Eq. 1 [↗](#). As for the solo session, we analyzed the proportion of variance in vigor explained by inter-dyad differences throughout the connected session. When grouping all the connected conditions, we found that intra-dyad (that is variability of vigor of a dyad across blocks) variability explained 63% of the total variance. However, when analyzing the blocks with and without viscous load separately, inter-dyad variability explained 99.4% (KL and KH) and 72.3% (in the four loaded conditions) of the variance. This means that the viscous loads changed how vigor scores are spread, thereby increasing intra-individual variability. However, when comparing conditions with similar dynamics, the variability in dyadic vigor scores was primarily accounted for by differences between dyads, as for lone movements. Therefore, each dyad seemed to behave as a single unit with its own preferred vigor.

To confirm this analysis, we then examined whether the relative dyadic vigor scores across the dyads changed with load and connection stiffness using Pearson correlation coefficients between pairs of conditions. All 15 Pearson correlations were significant ($p < 4.6 \cdot 10^{-3}$, Pearson- $r > 0.64$ in all cases), showing that the order of dyads according to their vigor scores was robust to evolving dynamics and effort levels (see Fig. 3I [↗](#) and Supplementary Fig. S.6J for examples of correlations). This confirms that vigor scores mainly vary across dyads rather than between conditions.

Therefore, we conclude that goal-oriented movements of mechanically connected individuals are characterized by a *dyadic vigor*, supported by: (i) coordinated motor plans between partners, evidenced by minimal interaction effort and movement duration differences; (ii) kinematics similar to individual movements, with overall higher performance (time, smoothness and accuracy); (iii) affine amplitude-duration relationships; (iv) larger inter-dyad than intra-dyad variability; and (v) vigor scores robustly correlated throughout all conditions.

Emergence of dyadic vigor from the partners' individual vigor

Connected movements durations were either shorter than those of the fast partners, in the unloaded conditions, or close to them in the loaded conditions (Fig. 3H [↗](#) and Supplementary Fig. S.6G–I). This suggests that the faster partner primarily determined the dyadic vigor, effectively acting as a leader. Such behavior rules out the *leader-follower* hypothesis where the slower

partner would lead. Conversely, the *leader-follower* hypothesis assuming the fast partner as leader, where the slow individual merely aligns their motor plan based on interaction forces, appears plausible but fails to explain the observed increase of movement velocity in unloaded conditions.

An alternative interpretation is that the fast partner indeed leads but also adjusts for the slower partner's vigor when setting the dyadic vigor, for instance via a convex combination with a higher weight on their own vigor. According to this *weighted adaptation* hypothesis, the dyadic vigor v^d would depend on the vigors v^f and v^s of the fast and slow partners as:

$$v^d = \alpha v^f + (1 - \alpha) v^s \quad (2)$$

where α allows to adjust the importance given to each partner: $\alpha = 1$ and $\alpha = 0$ fall back to the *leader-follower* hypothesis with a fast and a slow leader respectively, while $\alpha = 0.5$ represents equal contribution. Values $\alpha \in (0.5, 1)$ indicate a dominant fast leader marginally accounting for the slow partner's vigor, and conversely for $\alpha \in (0, 0.5)$. Based on the movement duration results, α would likely be close to 1 in our dataset.

To evaluate this hypothesis, we fitted a linear mixed model (LMM) assessing how dyadic vigor depends on the vigor of the fast and slow partners across conditions (see Eq. 5 in Methods). Surprisingly, dyadic vigor was significantly related to the fast partners' vigor only in KLV ($r = 0.14$, $p = 0.03$) and not in any other condition (KL & KH: $r < 0.14$, $p > 0.22$; KHV: $r = 0.13$, $p = 0.15$; KLVH & KHVH: $r < 0.1$, $p > 0.24$). In contrast, the slow partners' vigor significantly predicted dyadic vigor in five out of six conditions (KL & KH: $r > 0.624$, $p < 0.001$; KLV & KHV: $r > 0.24$, $p < 0.009$; KLVH: $r = 0.2$, $p = 0.028$).

This unexpected result invalidates the *leader-follower* hypothesis, which predicts that dyadic vigor should primarily depend on the fast partner's vigor. Moreover, the LMM analysis also indicates that dyadic vigor is not influenced by mixed parameters, such as the difference in vigor between partners or a convex combination of their individual vigors as in Eq. 2. In that case, the fast partner's vigor would have been a significant predictor. In conclusion, this refutes the *weighted adaptation* hypothesis, which assumes similar contributions from both partners to the dyadic vigor.

We are thus left with the *interactive adaptation* hypothesis, according to which both participants adapt their control strategy to the new dynamical context induced by the interaction with the partner. While individuals may not perfectly infer their partner's motor plan, they may identify the distribution of the partner's movement durations and plan their motion accordingly. In particular, each participant may partially anticipate the partner's timing and adjust their own vigor to minimize interaction efforts variability, which otherwise lead to instability. The effect of uncertainty induced by the partner's timing, accumulating over time, may prompt individuals to move faster than initially planned—consistent with evidence that higher speeds mitigate the detrimental effects of uncertainty on performance [28]. This dynamic change of dyadic vigor could explain why dyads move even faster than the faster partner in unloaded conditions.

Computational modeling of coordination mechanism

To test this premise, we designed a computational model representing this dynamic adaptation related to the partner's uncertainty. Based on the above observations, it aims at predicting (i) minimal variation of interaction efforts, as dyads were shown to coordinate their movements to allow the emergence of dyadic vigor, (ii) dyads moving faster than either partners in solo (when there is no load), (iii) dyads' movement times close to those of the fast partner in solo in presence of load, (iv) a dyadic vigor that depends mainly on the vigor of the slow partner, and (v) small dyadic movement duration prediction errors across the 30 tested conditions (5 targets $\times$ 2 stiffness levels $\times$ 3 viscous loads).

We simulated a dyad that adapts its behavior relative to an inferred distribution of the slow partner's movements (see Methods for details), while integrating the costs associated with the fast partner during planning. Accordingly, the model assumes asymmetric roles for the two partners, where the fast partner costs are used by the dyad to plan the movements while the uncertainty arising from the interaction with the slow partner acts as a higher level constraint through the

interaction torque. Conceptually, this asymmetry reflects the role of the slow partner in setting the vigor ranking within the dyads, and the role of the fast partner in determining the average dyadic behavior.

First, we extract an optimal control model of effort and time characterizing the participants' average behavior from data of the solo session (as explained in the Methods) [15]. Using the identified cost function, we can predict the movement duration and mechanical work of the average lone participant in the VL and VH conditions based on their initial null-field behavior (see Supplementary Fig. S.2). This identified individual motor plan serves as a basis for simulating the interaction within dyads. Specifically, the effort cost identified for an average subject is used to build the model associated with the *interactive adaptation*, while the time cost is identified separately for the slow and fast groups (see Methods for values of fitted parameters). To model the interaction uncertainty, the slow partner is represented by a distribution of minimum jerk trajectories of stochastic duration $T_k^s = \xi_k$ with average μ_k and deviation σ_k for target k , based on their average vigor. The dyads' common plan is then computed through the minimization of the expectation of a cost function considering the common effort cost and the individual cost of time of the fast partner, as well as the variations of interaction torque $\dot{\tau}_k^i$:

$$C_k(u_k, T_k) = \mathbb{E} \left[V_k^f(u_k, T_k) + Q_\tau \int_0^{T_k} g(\dot{\tau}_k^i; \xi_k) dt \right] \quad (3)$$

where $\mathbb{E}$ is the expectation with respect to the slow partner's distribution, V_k^f the cost function including the common cost of effort and the cost of time of the fast participant (see Eq. 10 in Methods), Q_τ a weight and T_k the free final time minimizing a positive function $g(\dot{\tau}_k^i; \xi_k)$ of the interaction torque variation $\dot{\tau}_k^i$, associated with the distribution of the slow partner's movements duration. The computation and optimization of the cost function are described in the Methods.

The model's predictions, summarized in Fig. 4, closely match the experimental data in both the bell-shaped, smooth velocity profile and the predicted movement durations. The median prediction errors across the 30 simulated conditions were small (KL: 28.7 ms; KH: 50.8 ms) with most errors below 80 ms except for A_1 with viscosity, A_2 with KL and viscosity, and A_3 – A_5 for KH without viscosity. Importantly, all these predictions were obtained using the same weighting Q_τ in the cost function in Eq. 3 (or Eq. 17 in the Methods for details). To summarize the evaluation of plausible hypotheses for the mechanism of coordination within a dyad, Table 1 presents a systematic comparison of the four tested hypotheses on their ability to reproduce both the qualitative and quantitative experimental findings. This analysis further supports the validity and relevance of the *interactive adaptation* hypothesis, and its implementation as an asymmetric model where the slow partner acts as an overarching constraint while the fast partner acts on the expected dyad behavior, compared with the alternative explanations tested.

How does the *interactive adaptation* model work? Co-adaptation of the two partners yields an average dyadic movement duration close to the average preferred duration of the fast group, but the slow partner ultimately determines which dyad is faster or slower relative to this average behavior. In our approach, the uncertain term associated with the slow partners should be understood as an overarching constraint that conditions the strategy of the dyad, while the fast partner cost of time acts as a contributor to the expected dyad strategy. Conceptually and numerically as reported in the sensitivity analysis, this asymmetry corresponds to the role of the slow partners in setting the vigor ranking among the dyads and the role of the fast partner in setting the average dyadic behavior. Importantly, the uncertainty associated with the distribution of the slower partner's motor plan may explain the faster movement in KL and KH. Specifically, the partners would try to minimize the accumulation of errors in expected variations of interaction efforts to stabilize the interaction, resulting in faster movements as this cost naturally increases with time (see uncertainty term in Eq. 17 in Methods). In this context, the slower partner would only be willing to incorporate this acceleration up to a certain point depending on their individual vigor, thereby determining the final dyadic vigor. This would also explain why this global acceleration is not observed in presence of a viscous load. Although such efforts may

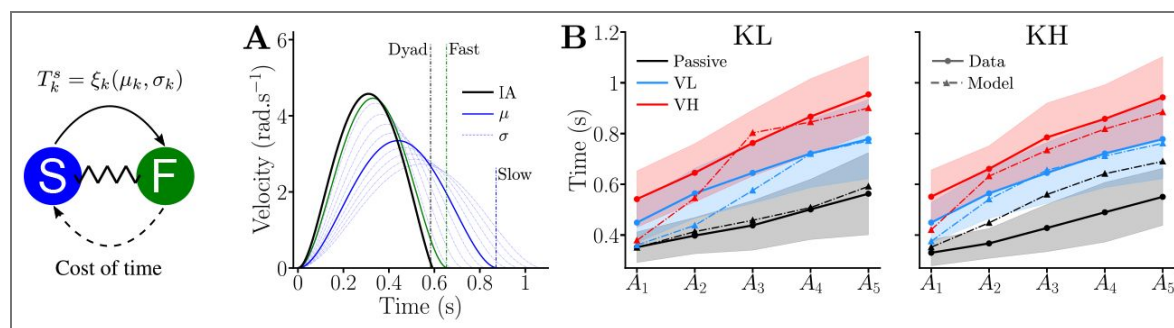
Table 1. Ability of the different models to reproduce the main qualitative (top) and quantitative (bottom) experimental findings.

The interactive adaptation hypothesis is the only one among those tested that can reproduce all the main qualitative experimental findings. It also results in the lower median prediction errors for the low stiffness conditions, and in the second lower median errors for the high stiffness movements, with a negligible 12 ms difference compared to the best predictions.

Characteristic	Co-activity	Leader-follower Fast lead	Leader-follower Slow lead	Weighted average	Interactive adaptation
Bell-shaped & smooth velocity profile	No	YES	YES	YES	YES
Low h-h interaction force	No	YES	YES	YES	YES
Dyad faster than fast partner	No	No	No	No	YES
Only slow partner impacts h-h vigor	YES	No	YES	No	YES
Median error low stiffness (ms)	138.2	35	138.2	(35,138.2)	28.7
Median error high stiffness (ms)	130.5	38	130.5	(38,130.5)	50.8

Figure 4. Predictions of dyad movement duration obtained with the model arising from the *interactive adaptation* hypothesis.

A. Illustration of predicted velocity profiles and measured averaged durations for movements towards A_5 in the KL condition. The solid and dashed-dotted blue lines represent respectively the average and samples of the distribution of the slow partner, with the average movement duration of the slow partners as a dashed dotted blue vertical line. The solid green line illustrates the average velocity profile of the fast partners moving alone, with the green dashed-dotted vertical line the corresponding duration in the experiment. Finally, the solid black represents the velocity profile predicted by an *interactive adaptation*, and the black dashed-dotted vertical line the average experimental duration of dyadic movements. **B.** Comparison between predicted movement time and data for all the connected conditions.



increase the deviation of the inferred partner's distribution by masking interaction efforts, they also naturally increase stability by dissipating energy. Consequently, they automatically reduce the variations in interaction efforts, and their associated detrimental effects on dyadic dynamics, thereby reducing the cost of increased movement duration. In our model, this important property naturally emerges from the compromise between time and effort, as identified for individual behavior, and uncertainty, due to the interaction with the partner, when controlling the dyad's dynamics.

Supplementary Figs. S.18,S.19 present the results of a sensitivity analysis varying the model's three parameters: weight of interaction torque variation minimization and the slow partner's movement time distribution average and deviation $T_k^s = \xi_k(\mu_k, \sigma_k)$. This analysis shows that (i) the model remains stable beyond the durations and parameters examined in the present study, and (ii) dyadic movement duration decreases as μ_k decreases for large ranges of parameters, consistent with the observed correlation between dyadic vigor and the slow participant's vigor. We also verified the effect of changing the cost of time of the fast partner, i.e. their vigor, on predicted movement duration (see Supplementary Fig. S.20). This analysis shows that modifying the fast partner's vigor only marginally impacts the optimal predicted time, with changes of 10 ms and 35 ms for a 30% increase in their cost of time for the A_1 and A_5 targets respectively. These minor impacts are consistent with the fast partner's vigor not being a predictor of dyadic vigor. The *interactive adaptation* is therefore the only hypothesis that captures all the qualitative features of the experimental data, and the associated model provides accurate movement duration predictions, thereby complying with all the requested conditions for computational modeling summarized above. Furthermore, as the impact of the fast partner's vigor on the predicted time is small, the predicted motion plan can be considered as representing the dyad's emerging common plan, rather than a plan that would result from a fast leader that would just account for the slow partner's behavior.

2.3 After effects of dyadic interaction

Did dyadic practice influence individual movement timing? To address this question we analyzed how the movements of slow and faster partners changed with dyadic practice. Fig. 5A [↗](#) shows that averaged trajectories across the population were notably different after (NF2) relative to before (NF1) practice. In particular, the average movement durations decreased, while the velocity profiles remained similar to those with both KL and KH, without terminal submovement. Furthermore, the average amplitude-duration relationship in NF2 changed for the slow partners but not for the fast (Fig. 5B [↗](#)).

A one-way ANOVA confirmed behavioral differences between groups and conditions ($SS > 0.3$, $p < 10^{-3}$, $\eta^2 > 0.08$ in both cases). In the first null-field block (NF1), the slow partners moved significantly slower than the fast partners ($p < 10^{-3}$, Cohen's $D = 0.72$) but not in the post-connection null-field block (NF2) ($p > 0.72$). Furthermore, the group of slow partners changed their movement durations between NF1 and NF2 ($p < 10^{-3}$, Cohen's $D = 0.8$) but not the group of fast partners ($p = 0.41$). Movement amplitude had a main effect on time saving ($W = 0.32$, $p < 10^{-3}$), which was confirmed across the population for targets $\{A_4, A_5\}$ ($p < 0.02$, Cohen's $D > 0.87$ in all cases, Fig. 5C [↗](#)). The vigor analysis (Fig. 5D [↗](#), Supplementary Fig. S.21A) confirmed this adaptation: while the slow partners' vigor increased between NF1 and NF2 ($p = 0.01$, Cohen's $D = 2.73$), the fast partners' vigor did not change ($p = 0.43$).

Interestingly, despite the change of motor plan of the slow partners, the NF1 and NF2 vigor scores over the whole population remained correlated ($p = 3.2 \cdot 10^{-3}$, $r = 0.62$, see Supplementary Fig. S.21B). Furthermore, the vigors of the group of slow partners in NF2 were correlated to their behavior in NF1 ($p = 0.004$, $r = 0.87$). Surprisingly, the vigors of the group of fast partners in NF2 was correlated with the vigor of their slower partner in NF1 ($p = 0.034$, $r = 0.73$), but not with their own vigors in NF1 ($p = 0.14$, $r = 0.5$). This highlights a structural change in the repartition of vigor scores in the fast partners, reflecting their adaptation to the slow partners in each dyad.

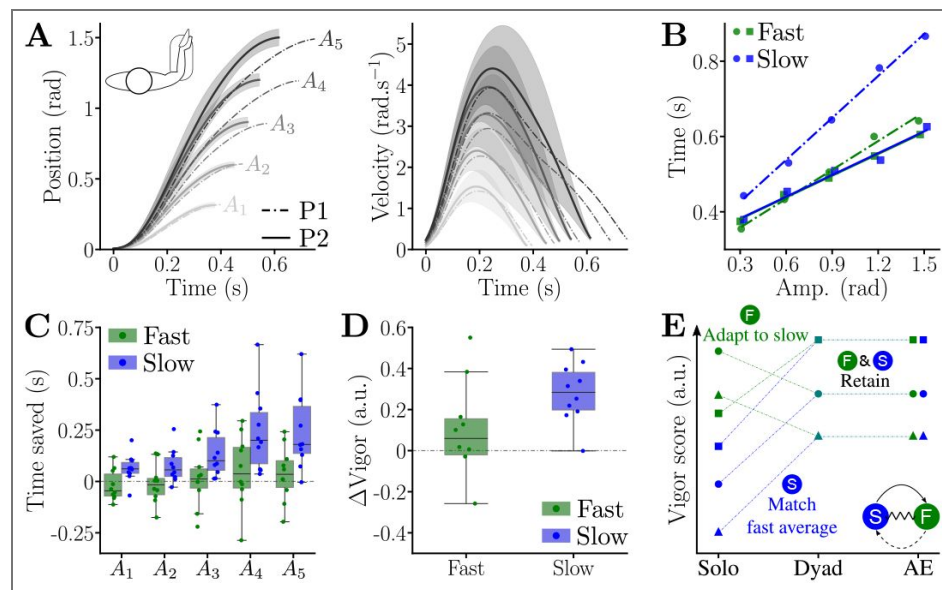


Figure 5. Kinematics and vigor before and after human-human interaction.

Data before (NF1) and after (NF2) the dyadic session are represented using dashed-dotted and solid lines, respectively. **A.** Population average of the position and velocity for each target. **B.** Averaged amplitude-duration relationships for the fast and slow partners in NF1 and NF2. **C.** Time saved between NF1 and NF2 for the fast and slow partners and for each target. **D.** Difference in vigor scores between NF1 and NF2, for the fast and slow partners. **E.** Illustration of the fast and slow partners' adaptations. When connected, the average vigor of the slow partners increases to match the one of the fast partners, while the fast partners' group changes its internal ranking in vigor to match the ranking of the slow partners. These adaptations are retained for both groups in after-effects (AE).

Therefore, dyadic practice led to an adaptation of the average vigor in the slow partners, but their relative repartition was not modified. That is, fast participants within the group of slow partners remained fast after dyadic practice, and slow participants remained slow. In contrast, the average vigor scores of the fast partners remained unchanged in NF2 relative to NF1. However, there was a structural change in their repartition according to vigor, which matched the repartition of their slow partner after dyadic practice (Fig. 5E). Importantly, at the dyad level the adaptation is sometimes larger for the slow participant and sometimes larger for the fast participant. For instance, the dyad composed of subjects S5 and S6 shows a large decrease of vigor for the fast (S6) and an increase of a lower magnitude for the slow (S5) participant after connected practice (see Supplementary Fig. S.21 and Supplementary Table S.1 for details).

3 Discussion

The movements of individuals are characterized by their vigor, revealing how prone individuals are to perform actions fast or slowly in various conditions. Vigor is thought to be a fundamental idiosyncratic trait of volitional movements that relates to energy, time/reward, and performance/accuracy. How do individuals combine their vigor to coordinate movements when physically collaborating on joint actions? We addressed this question by investigating how partners connected by a virtual elastic band performed goal-directed hand movements without time constraints. Unlike previous human interaction studies, where timing was externally prescribed, either by a tracking task [19,20,29], explicit instructions [17], or the introduction of an explicit reward when connected [16], our study allowed for natural and unbiased timing emergence.

We first observed that connected partners used a minimal interaction force to achieve synchronization, rapidly adapting within the initial trials, and exhibiting negligible time differences in movement execution. This indicates that partners did not move independently, as proposed by [17], but instead rapidly adopted a common motor plan, likely facilitated by humans routinely interacting in daily tasks and the studies task simplicity. Importantly, these findings show that sensory exchanges occur between partners even during simple point-to-point movements, not only in complex interaction scenarios as previously reported [29–32]. Second, in conditions without viscous load, dyadic movements were strikingly faster and smoother than the movements of either of their individuals, without compromising accuracy. With a load the group of slower partners moved as fast as the faster partners, while the group of faster partners essentially maintained their pace. These findings confirm and extend the benefits of dyadic control, in terms of movement time and trajectory tracking accuracy previously observed in collaborative actions [16, 21, 29].

Furthermore, dyads exhibited consistently slower or faster movement duration relative to the average over dyads, which was shown for different levels of effort and connection stiffness. As dyads' partners coordinated their actions with these systematic features, this allowed us to define *dyadic vigor* similarly to individual vigor [12, 23]. Interestingly, we found no significant differences in reaction times between the groups of fast and slow partners, in contrast to previous works reporting a correlation between reaction time and vigor [33]. This discrepancy may stem from several factors. First, our task was not specifically designed to finely assess reaction times, e.g. using eye gaze motion initiation instead of hand motion initiation. Second, the correlation between these two parameters may be weak, as suggested by non-significant trends in previous works [3, 24]. Interestingly, we observed that dyadic practice led to significantly shorter reaction times, which may be due to increased attention [34], possibly related with task uncertainty, when one's motor behavior is directly impacted by a partner. This finding may have important implications for the design of robotic partners physically interacting with humans, where being synchronized with the user is critical.

We found a striking result regarding this dyadic vigor, which we expected to primarily correlate with the fast partners' vigor in a leader-follower scheme [18]. While the connected participants had a movement duration close to that of the group of fast partners, dyadic vigor was instead predicted by the slower partner's vigor. This unexpected finding was consistently observed across

varying stiffness and resistance conditions, where the slower partners' vigor was a significant predictor of dyadic vigor. This reorganization of the fast partners' vigor scores highlights the slow partners' pivotal role in determining the dyad's movement pace. While the slower partners move faster connected than alone, the faster partners adapt their effort accordingly to match the slow partners' influence. Note that the magnitude of the adaptation can be large for both of them, sometimes larger for the faster.

This finding challenges the leader-follower framework often used to describe human-human and human-robot interactions, where the leader would typically be the individual exerting greater motor force or being ahead in the movement [16, 19, 20]. Our results show that this interpretation is misleading. It also challenges the weighted averaging hypothesis suggesting that the dyad vigor should be systematically in-between the slow and fast partners' initial vigor, with both contributing to its setting. Rather, our results demonstrate that a dyad motor plan emerges from a co-adaptation where the slower partner's vigor is the final determinant. Instead of a fixed leader-follower relationship or weighted adaptation, human-human interactions appear to depend on the context (e.g. the available information to each partner, or the type of task) rather than on predefined roles or a simple mix of the initial individual strategies [35]. This should be considered when designing human-robot interaction controllers [36], contrary to the numerous control methods relying on a leader-follower strategy to provide assistance [18].

Following connected training, strong after effects were observed in both fast and slow partners. For slow partners, a large increase in the individual vigor was observed. Interestingly, the vigor scores of the fast partners were reorganized depending on their partner, while the order of the slow partners' scores was unchanged by dyadic practice, similarly to dyadic vigor during the connected session. This highlights how both partners adapted. The slow partners increased vigor, converging to the average vigor of the fast partners, while the fast partners reorganized their vigor score corresponding to the repartition of the slow partners. These effects persisted across 100 simple reaching movements, whereas after-effects of force fields or visuomotor rotations are typically re-optimized within a few trials [37, 38], and effects of previous attempts at modifying individual vigor tended to vanish through unconstrained practice [39]. This persistence indicates a potential change in the time perception for both groups after practicing with a partner.

These findings suggest joint training as a potential therapeutic approach to restore declining movement vigor, such as in individuals with Parkinson's disease [40, 41]. In this context, future works should examine whether the vigor changes observed after dyadic practice are retained over longer periods or with intensive movement sessions, or whether they end up vanishing as was observed in studies where participants were instructed to move faster using visual cues [39]. If the vigor adaptations observed during and after physical human-human interactions such as in the present paper were shown to persist over long periods, this would challenge the view that vigor is a strictly stable, idiosyncratic trait. Instead, it would suggest that the exploration of other motor solutions and rewards can modify the optimal tradeoff selected by the CNS. Such fundamental changes in vigor could potentially be assessed by analyzing the evolution dopamine levels in the basal ganglia [42, 43] (e.g. using positron emission tomography [44]), as dopamine plays a major role in vigor regulation [7, 8, 10, 11, 45, 46].

To interpret the mechanism behind the coordinated movement timing and trajectory in dyads, we developed a computational model based on stochastic nonlinear optimal control with free final time, thereby combining several computational approaches [2, 3, 47], in which dyads plan movements based on the slower partner's average vigor under uncertainty, as was observed during connected movements. This model successfully predicts the key qualitative features of our data, in contrast to hypotheses that assume i) independent motor plans for slow and fast partners (*co-activity*), ii) a vigor distribution decided by a leader and accepted by a follower (*leader-follower*) or iii) a negotiation based on a weighting of their initial strategies in the task (*weighted adaptation*). These results further show the role of the slow partner and the critical impact of uncertainty with respect to the partner's plan in the emergence of dyadic vigor. These findings argue in favor of a dynamic adaptation of an individual's motor strategy caused by the presence of the other partner (and its inherent induced uncertainty).

Focusing on dyadic movement timing, the proposed model necessarily involved several simplifying assumptions. First, because current control theory does not, to our knowledge, yet allow the simulation of multiple physically interacting agents with nonlinear and uncertain dynamics under free-final-time conditions, dyadic interaction was approximated using a single-agent formulation. Second, we modeled the *average* dyad rather than generating dyad-specific predictions, as individual model fitting was unlikely to yield additional insights into the underlying coordination principles. We further restricted the model to the open-loop component of motor control, consistent with the stable, reproducible behavior observed after an initial increase in interaction force. As we predicted averaged movement durations across the experimental population for each condition, the influence of feedback control [25] was minimal. Extending the model with feedback control would require exploring the interplay between feedback control and (re)planning mechanisms that are beyond the scope of the present study, which primarily aims to investigate the role of individual vigor in shaping human-human interactions.

More fundamentally, vigor is thought to emerge from the interplay between effort, accuracy, and time or reward. Inter-individual differences in vigor may therefore reflect differences in sensitivity to any of these factors [6]. However, disentangling these sensitivities based solely on behavioral data is difficult. Here, we estimated the cost of time for the average participant using a previously established deterministic effort model with fixed final time [15]. Because the model does not explicitly account for accuracy (e.g. through variance minimization), its generalizability to tasks with different accuracy constraints is limited.

Our model achieved minimal movement duration prediction error across all 30 tested conditions, covering five amplitudes and six load conditions. Furthermore, we conservatively assumed that participants used a single cost function across all conditions, even though humans can adjust strategies based on task demands [48]. A sensitivity analysis confirmed that increasing the slow partner's average movement duration leads to an increase in the optimal dyadic movement duration, while changes in the fast partner's cost of time had marginal influence, consistent with the observed correlation between slower partners vigor and dyadic vigor.

These results indicate that the faster individual in a dyad relies on an internal estimate of the partner's movements distribution, providing a plausible mechanism for the faster dyadic movements in the absence viscous load, the preserved or decreased average movement duration of the fast partners in presence of viscous load, and the dependence of dyadic vigor on the slower partner. This model offers a principled basis for developing robotic systems that interact with humans similar to a human partner, with applications in collaborative robots for manufacturing [49], rehabilitation robotics [50], exoskeletons [51] and supernumerary robotic limbs [52].

4 Methods

4.1 Participants and experimental setup

The experimental protocol was approved by the Imperial College's Science, Engineering and Technology Research Ethics Committee (SETREC number 7112072). $N = 20$ right-handed young adults (4 females) with no known motor or cognitive impairment participated in the experiment in 10 dyads of two partners. Their biographical data were: age 27.5 ± 5.5 years old, weight 73.7 ± 14.5 kg, height 175 ± 8.5 cm and right hand length 19.1 ± 1.2 cm (from the wrist flexion/extension rotation axis to the farthest fingertip). Each participant was informed about the experiment, and signed an informed consent form before participating in it.

Each of the dyad's partner interacted with an HRX-1 robotic interface (HumanRobotiX, London, UK) using their right wrists, while seated in front of their respective monitor (Fig. 1B). Participants had to control a 0.6 cm diameter cursor towards a 2.7×2.7 cm² target via wrist flexion/extension. These dimensions ensured consistent movement amplitudes across participants with minimal accuracy demands. The two HRX-1 were controlled and synchronized at 1 kHz using a TI PiccoloF28069M LaunchPad. Wrist flexion/extension angles were recorded at 300 Hz with an

optical encoder (MILE 512-6400, 6400 counts per turn). Angles (18° , 36° , 54° , 72° , 90°) were linearly mapped to targets (A_1 , A_2 , A_3 , A_4 , A_5) on the monitor, with 9.3 cm between the centers of consecutive targets (Fig. 1B [↗](#)).

4.2 Experimental protocols

Participants completed two experimental sessions (Fig. 1C [↗](#)):

- In a first *solo session*, the two participants performed goal-directed wrist flexion/extension movements independently. These movements were used to: (i) verify that our population sample followed a vigor law, (ii) estimate their individual vigor and cost of time for computational modeling, and (iii) establish a baseline to analyze after-effects of the human-human interaction.
- In the subsequent *dyadic session*, they were coupled via a virtual elastic band implemented through the robotic interfaces. The data enabled us to investigate movement coordination between the partners and its dependence on individual vigor.

The participants performed back and forth reaching at preferred speed, starting with a wrist extension movement from A_0 to one of the targets A_1 , A_2 , A_3 , A_4 , A_5 , followed by a flexion movement, back to A_0 . After each movement, the timing of the next movement start was drawn from a uniform distribution to prevent anticipation and rhythmic behavior. After holding at A_0 for 2 ± 0.15 s, A_0 was switched off and a target was switched on for 2 ± 0.15 s, then A_0 was switched on again. This cycle was then repeated.

Solo session

Participants completed 3 blocks of 100 trials (10 extension and 10 flexion movements per amplitude: $\{18^\circ, 36^\circ, 54^\circ, 72^\circ, 90^\circ\}$) in a fixed pseudo-random sequence. A two-minute break was provided between blocks to prevent fatigue. The first block was performed in “transparent” mode (robot motor off). The next two blocks introduced resistive viscous loads ($\nu \in \{0.075, 0.15\}$ Nm s/rad in random order) to examine vigor robustness under varying effort costs. A final washout block was introduced to avoid carrying after-effects of the two visco-resistive blocks into the connected session. This block was shorter than the other blocks, consisting of 40 reaching movements with 4 flexions and 4 extensions per amplitude. The same amplitudes and targets were used as for the other blocks but their order was not randomized, i.e. the blocks consisted of 4 series of movements towards A_1 , then A_2 , ..., then A_5 .

Dyadic session

Dyads of participants completed 6 blocks of 100 trials while coupled via a virtual visco-elastic band. The first two blocks used an elastic band of stiffness $\kappa \in \{0.5, 1.6\}$ Nm/rad in randomized order. The next four blocks tested all κ and ν combinations in random order to explore time-effort tradeoffs. A seventh block without coupling (i.e. $\kappa = \nu \triangleq 0$) was used to assess carry over effects of the interaction with the partner on individual vigor. Participants were informed that the robot would provide assistance but were not told that they were interacting with another human. When debriefed at the end of the experiment, only one of the 20 participants reported having realized that they were connected to a partner. Most participants believed that they were interacting with a version of themselves or with a robot exhibiting stochastic behavior.

4.3 Data processing

Wrist kinematics

The wrist velocity and acceleration were computed by numerical differentiation, after low pass filtering of position data using a forward-backward fifth order Butterworth filter with 5 Hz cut-off frequency. Wrist flexion and extension movements were first grossly segmented based on the timings of the targets, then a threshold of 5% of peak velocity was used to compute the start and end of each movement. Leftwards and rightwards movements were pooled together as their vigor was similar, consistently with expectations [[3](#), [23](#)].

The amplitude-duration relationship of each participant was obtained by i) averaging movement durations and amplitudes per target, ii) performing an affine least-square regression (using the *LinearRegression* function of *scikit-learn.linear_model*). The amplitude-duration relationship of a group of participants was obtained by: i) averaging across the group the average movements durations and amplitudes of each participant, for each target separately, then ii) performing an affine regression.

All the movement durations were obtained from affine fittings of the amplitude-duration relationships. The vigor scores in the VL and VH conditions were computed using the average movement duration of the population for each target $\bar{T}(A_k)$ from NF1, which allowed to illustrate the general decrease in vigor with a resistive torque. The same approach was adopted for the dyadic session, using the population averaged movement duration in KL and KH to compute the vigor scores in {KLV L, KLV H} and {KHV L, KHV H}, respectively. As a consequence, the vigor scores were equally distributed around 1 only in the NF1, KL, and KH conditions. To analyze the variance, the vigor scores were computed based on the average movement duration of the population in each condition.

The reaction time of participants was computed as the time difference between the apparition of a target and the first instant where the wrist angular velocity exceeded 0.01 rad/s.

Interaction forces

The reported average values were obtained by (i) computing the absolute value of the interaction efforts as a function of time for each whole movement, and (ii) computing the average value of this absolute profile.

4.4 Statistical analysis

The statistical analyses were performed using custom Python 3.8 scripts, the *statsmodel* package [53], and the *Pingouin* package [54]. For all metrics, the results were first averaged in each participant, condition, and movement amplitude before performing statistical analyses. Normality (*Shapiro-Wilk* test [55]) and sphericity (*Mauchly's* test [56]) of the distribution of the residuals were not always verified. Therefore, for convenience, all within subjects and within dyads main effects of the condition and amplitude were assessed using Friedman tests, while unpaired effects between groups were assessed using Kruskal-Wallis tests, with a significance level set at 5%.

In case of a significant main effect, post-hoc pairwise comparisons were performed using Wilcoxon tests within subjects, and Mann-Whitney tests between the fast and slow partners. The significance level of post-hoc comparisons was set at 5%, with a Bonferroni-Holm correction for multiple comparisons. All the significant comparisons are reported with the Cohen's D value to illustrate the effect size, where $D < 0.4$ would be considered as small and reported as a weak result.

The proportion of variance due to inter-individual variability was quantified using a variance decomposition of the recorded movement durations data. Sum-of-square components were used to compute the percentage of variance explained by inter-individual variability as follows:

$$\text{Var}_{\text{inter}} = 100 \frac{\text{SS}_{\text{inter}}}{\text{SS}_{\text{inter}} + \text{SS}_{\text{intra}}}. \quad (4)$$

The prediction of the dyads' vigor in the connected conditions was performed using a linear mixed model attempting to correlate the vigor of the dyad to the vigor of its fastest and slowest participants as

$$v^d \sim \gamma^f v^f + \gamma^s v^s, \quad (5)$$

where v^d is the vigor of dyads in one of the connected conditions {KL, KLV L, KLV H, KH, KHV L, KHV H}, v^f is the vigor of the dyad's faster partner in the NF1 condition and v^s of the slower partner. Throughout the paper a Bonferroni-Holm correction was applied for multiple regressions.

4.5 Computational modeling and simulations

4.5.1 Solo session

Data collected in solo setting were used to model the dyad behavior and predict the dyadic behavior from individual movement characteristics. The solo session included three levels of viscous resistance $\nu \in \{0, 0.075, 0.15\}$ Nm s/rad (referred to as NF1, VL and VH, respectively), resulting in the task dynamics

$$I\ddot{q} = \tau - D\dot{q} - \nu\dot{q}, \quad (6)$$

where $\ddot{q}$ [rad/s²] is the wrist angular acceleration, $\dot{q}$ [rad/s] the angular velocity, τ [Nm] the human wrist torque, $I = 9.2 \cdot 10^{-3}$ Nm s² the population average hand+robot inertia, and $D = 0.03$ Nm s is the wrist damping selected in the range of values identified during null-field movements in [57, 58].

Following the inverse optimal control procedure of [3, 15], we first estimated the cost of time of the average participant based on the population average amplitude-duration relationship (see Fig. 2B) in NF1 without viscous load. The estimated amplitude-duration relationship for the population was

$$\bar{T}(A_k) = 0.314 A_k + 0.298, \quad A_k \in \{18^\circ, 36^\circ, 54^\circ, 72^\circ, 90^\circ\}. \quad (7)$$

Moving the hand from an initial angle q_0 to and end angle q_e was formulated as an optimal control problem, using the minimum torque change framework [59]. Utilizing the state $\mathbf{x}_k = (q_k, \dot{q}_k, \tau_k)$ and the motor command $u_k = \dot{\tau}_k$, the cost of effort to minimize for amplitude k was

$$E_k(u_k) = \int_0^{\bar{T}(A_k)} [\mathbf{x}_e - \mathbf{x}_k(t)]' Q_k [\mathbf{x}_e - \mathbf{x}_k(t)] + \beta u_k^2 dt \quad (8)$$

where $\mathbf{x}_e = [q_e, 0]'$ is the final state, $\bar{T}(A_k)$ is the final time of the average participant for amplitude k computed from Eq.(7). $Q_k = \text{diag}(15.5/A_k, 0, 0)$ and $\beta = 0.95$, determined by trial and error, allow to adjust the compromise between minimizing the torque change and the distance to the target. As detailed in [15], the human cost of time $G(t)$ can be identified by fitting and integrating a sigmoid curve on the Hamiltonians of the fixed final time problems from Eq. 8, yielding

$$G(t) = p_1 (1 - [1 + (t/p_3)^{p_2}]^{-p_4}), \quad (9)$$

where $\mathbf{p} = (p_1, p_2, p_3, p_4) = (1.5713, 6.9723, 0.4497, 1)$ was identified for the average participant in our sample. Using the same effort formulation as in Eq. 8, it is then possible to predict the movement duration and cost of effort for $\nu \in \{0.075, 0.15\}$ Nm s/rad by minimizing

$$V_k(u_k, T_k) = G(T_k) + \int_0^{T_k} [\mathbf{x}_e - \mathbf{x}_k(t)]' Q_k [\mathbf{x}_e - \mathbf{x}_k(t)] + \beta u_k^2 dt \quad (10)$$

in free final time, i.e. when the movement duration is optimized simultaneously to the effort cost. Here V_k is the cost to minimize for amplitude A_k and $T_k \in [0.01, 10]$ s the optimal final time for amplitude k . Note that minimizing V_k with $\nu \triangleq 0$ leads to $T_k = \bar{T}(A_k)$, and to the same predicted motor behaviors as the fixed final time optimal control problems minimizing Eq. 8.

The simulations of the solo session were performed using the Matlab (R2021a, MathWorks) version of GPOPS-II [60–62], which is based on an orthogonal collocation method relying on SNOPT to solve the nonlinear programming problem [63].

4.5.2 Dyadic session

The task of the dyadic session included two levels of connection stiffness $\kappa \in \{0.5, 1.6\}$ Nm/rad, and three levels of viscous resistance as for the solo session, yielding the coupled dynamics of the fast and slow partners

$$\begin{cases} I \ddot{q}^f = \tau^f - D \dot{q}^f - \kappa(q^f - q^s) - \nu \dot{q}^f \\ I \ddot{q}^s = \tau^s - D \dot{q}^s - \kappa(q^s - q^f) - \nu \dot{q}^s \end{cases} \quad (11)$$

where the superscripts f and s indicate the fast and slow participants, respectively, τ^f and τ^s are their wrist torques, I is the total inertia of the wrist+exoskeleton, D the total damping as in the solo session, and $\ddot{q}$, $\dot{q}$, q are the wrist angular acceleration, speed and position, respectively.

To simulate movements according to the *co-activity* hypothesis, we split the participants in the groups of the faster and slower partners in each dyad according to their vigor in NF1. Furthermore, to simulate the *interactive adaptation* hypothesis, assuming a coordination of the partners' motor plans, we had to identify the cost of time of the fast and slow partners, which was done based on their respective averaged amplitude-duration relationships

$$\begin{cases} \bar{T}^f(A_k) = 0.235A_k + 0.285 \\ \bar{T}^s(A_k) = 0.35A_k + 0.323 \end{cases} \quad (12)$$

Importantly, to simulate *interactive adaptation* we identified the cost of time of the fast ($\mathbf{p} = (3.9615, 5.8958, 0.4459, 1)$) and the slow partners ($\mathbf{p} = (1.0184 \cdot 10^7, 11.2958, 1.9980, 1.4447 \cdot 10^{-5})$) separately, which allowed us to simulate the effects of the average partners' individual vigor on the dyad's motor plan (see Eq. 9 for the definition). Then, the simulated hypotheses induced different state variables, conditions and numerical methods to predict movements under the coupled dynamics as described below.

Co-activity

assumes that the participants do not coordinate their movements, with each partner trying to perform their predefined desired trajectory (as suggested in [17]). For the sake of simplicity, we set these predefined plans using simple minimum jerk plans with durations identified in the lone session for the fast and slow participant separately. Optimal control commands $u_k^f(t)$ and $u_k^s(t)$ to track the obtained desired trajectories for the fast and slow partners for target k , respectively, are computed using a linear quadratic regulator. We use the state $\mathbf{x} = (q_k^f, \dot{q}_k^f, q_k^s, \dot{q}_k^s)$ and the cost matrices $\mathbf{Q} = \text{diag}(100, 0.1, 100, 0.1)$ and $\mathbf{R} = \mathbb{I}_2$. The fast partners' trajectory is extended with null values to match the duration of the slow partners' plan. The resulting coupled movements are Euler integrated using these control inputs in Eq. 11.

Interactive adaptation

Here we assume that dyads' partners collaboratively optimize their movements based on the solo behavior of the slow partner, who "drives" the co-adaptation. Our model describes the dyad's motion plan using a single cost function. While one would intuitively seek for multiple agents modeling, to our knowledge there is no tractable solution to simulate multiple agents under uncertainty, in free final time, and not based on usual quadratic cost functions. Both partners contribute to selecting the final movement control by optimizing the common motor plan based on a prediction of the slow partner's motion, which is uncertain for the fast partner. For the sake of simplicity, we represent the slow partner's trajectory as a distribution of minimal jerk trajectories [64]

$$q_k^s(\xi_k) = q_0 + A_k \zeta^3(10 - 15\zeta + 6\zeta^2), \quad \zeta \equiv \frac{t}{\xi_k} \quad (13)$$

where the movement duration $T_k^s = \xi_k$ is a random variable with mean μ_k and standard deviation σ_k . Note that q_0 and the $\{A_k\}$ are considered common and known to both partners, and no assumption is made on ξ_k 's distribution beyond its deviation. The task dynamics inferred by the fast partner are

$$I \ddot{q}_k^f = \tau_k^f - (D + \nu) \dot{q}_k^f - \tau_k^i, \quad \tau_k^i = \kappa[q_k^f - q_k^s(\xi_k)], \quad (14)$$

where τ_k^i is the interaction torque for target k . Since ξ_k is a random variable, the uncertain interaction torque change can be approximated based on a linearization of the velocity of the slow partner around μ_k :

$$\dot{q}_k^s(\xi_k) = \dot{q}_k^s(\mu_k) + \left. \frac{\partial \dot{q}_k^s}{\partial \xi_k} \right|_{\mu_k} (\xi_k - \mu_k) \quad (15)$$

Yielding

$$\tau_k^i(\xi_k) = \kappa \left[\dot{q}_k^f - \dot{q}_k^s(\mu_k) \right] - \kappa \left. \frac{\partial \dot{q}_k^s}{\partial \xi_k} \right|_{\mu_k} (\xi_k - \mu_k). \quad (16)$$

The expected cost function of the dyad can be expressed as a tradeoff between the fast partner's cost function when moving alone (including their cost of time) and the uncertain interaction torque change with respect to the behavior of the slow participant, reflecting a strategy of stabilization of the interaction efforts. We consider that participants would minimize a cost related to the expected interaction torque and its uncertainty, while neglecting the cross-covariance terms, expressed as in the main text:

$$C_k(u_k, T_k) = \mathbb{E} \left[\underbrace{V_k^f(u_k, T_k)}_{\text{Alone cost fast}} + Q_\tau \int_0^{T_k} \left(\underbrace{\kappa^2 [\dot{q}_k^f - \dot{q}_k^s(\mu_k)]^2}_{\text{Expected interaction torque variations}} + \underbrace{\kappa^2 \left[\left. \frac{\partial \dot{q}_k^s}{\partial \xi_k} \right|_{\mu_k} (\xi_k - \mu_k) \right]^2}_{\text{Uncertainty term}} \right) dt \right] \quad (17)$$

This leads to the expected cost

$$C_k(u_k, T_k) = \bar{V}_k^f(u_k, T_k) + Q_\tau \kappa^2 \int_0^{T_k} \left(\left[\bar{\dot{q}}_k^f - \dot{q}_k^s(\mu_k) \right]^2 + P_q^f + \left[\left. \frac{\partial \dot{q}_k^s}{\partial \xi_k} \right|_{\mu_k} \right]^2 \sigma_k^2 \right) dt \quad (18)$$

where $\bar{V}_k^f = \mathbb{E}[V_k^f]$, u_k , u is the dyadic motor command, σ_k^2 is the variance of ξ and P_q^f the velocity component of the state covariance matrix. This stochastic optimization problem can be approximated under a deterministic form and solved within the stochastic optimal open-loop control framework of [47] (see in particular Eq. 3), which provided codes on which we based our simulations, using Julia [65], JuMP [66] and Ipopt [67], where the problem was transcribed into an NLP problem using a trapezoidal scheme [68].

In the simulation results of Fig. 4B corresponding to the *interactive adaptation* hypothesis, the dyads were considered to use the same cost function for all amplitudes and conditions with $Q_\tau = 3$. The uncertainty with respect to the slow partner's behavior was assumed to increase with movement amplitude and resistive viscosity. Specifically, we used values uniformly spread within $\sigma_k \in [0.15, 0.25]$ s for KL and KH, within $\sigma_k \in [0.2, 0.3]$ s for KLV and KHV, and within $\sigma_k \in [0.32, 0.4]$ s for KLVH and KHVH.

Data availability

All data supporting the findings of this study are available within the paper, the supplementary material, and are publicly available on Zenodo [69].

Acknowledgements

We thank Jingwen Zhao for her help during the development of the experimental setup, Claudia Clopath for her comments on a prior version of the manuscript and Dario Farina for his help in funding administration. This study was funded in part by the European Commission (FETOPEN H2020 899626 NIMA, ICT 871767 ReHyb and ERC NaturalBionics).

Additional information

Contribution statement

- Conceptualization – all authors
- Methodology – all authors
- Software – DV, BB
- Validation – DV
- Formal analysis – DV, BB
- Investigation – DV
- Resources – EB
- Data Curation – DV
- Visualization – DV, EB
- Supervision – EB
- Project administration – EB
- Funding acquisition – EB
- Writing - Original Draft – DV
- Writing - Review & Editing – all authors

Funding

Funder	Grant reference number	Author
EC European Research Council (ERC)	https://doi.org/10.3030/810346	Etienne Burdet
EC H2020 PRIORITY 'Excellent science' H2020 Future and Emerging Technologies (FET)	https://doi.org/10.3030/899626	Etienne Burdet
EC H2020 IL LEIT H2020 LEIT Information and Communication Technologies (ICT)	https://doi.org/10.3030/871767	Etienne Burdet

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Additional files

[Supplementary materials](#) 

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Peer reviews

Reviewer #1 (Public review):

Summary:

The authors present a novel investigation of movement vigor of individuals completing a synchronous extension-flexion task. Participants were placed into groups of two (so-called "dyads") and asked to complete shared movements (connected via a virtual loaded spring) to targets placed at varying amplitudes. The authors attempted to quantify what, if any, adjustments in movement vigor individual participants made during the dyadic movements, given the combined or co-dependent nature of the task. This is a novel, timely question of interest within the broader field of human sensorimotor control.

Participants from each dyad were labeled as "slow" (low vigor) or "fast" (high vigor), and their respective contributions to the combined movement metrics assessed. The authors presented four candidate models for dyad interactions: (a) independent motor plans (i.e., co-activity hypothesis), (b) individual-led motor plans (i.e., leader-follower hypothesis), (c) generalization to a weighted average motor plan (i.e., weighted adaptation hypothesis), and (d) an uncertainty-based model of dynamic partner-partner interaction (i.e., interactive adaptation hypothesis). The final model allowed for dynamic changes in individual motor plans (and therefore, movement vigor) based on partner-partner interactions and observations. After detailed observations of interaction torque and movement duration (or, vigor), the authors concluded that the interactive adaptation model provided the best explanation of human-human interaction during self-paced dyadic movements.

Strengths:

The experimental setup (simultaneous wrist extension-flexion movements) has been thoroughly vetted. The task was designed particularly well, with adequate block pseudo-randomization to ensure general validity of the results. The analyses of torque interaction, movement kinematics, and vigor are sound, as are the statistical measures used to assess significance. The authors structured the work via a helpful comparison of several candidate models of human-human interaction dynamics, and how well said models explained variance in the vigor of solo and combined movements.

The authors adequately addressed several concerns that I raised in my initial review of the work, including clarity regarding analyses of movement vigor and inclusion of additional analyses of reaction time. The results are supported by both parametric and non-parametric statistical methods.

The research question is timely and extends current neuroscientific understanding of sensorimotor control, particularly in social contexts. This work answers several new, important questions about control of vigor during volitional movements, and in doing so it motivates future research into the topic.

Weaknesses:

My chief concern about the study is the relatively low number of dyad data points ($n=10$). The authors recruited 20 participants, but the primary conclusions are based on dyad-specific interactions (i.e., analyses of "fast" vs "slow" participants in each pair). However, it is important to note that most of the effects upon which the conclusions rest are associated with relatively large effect sizes.

<https://doi.org/10.7554/eLife.109781.2.sa3>

Reviewer #2 (Public review):

Summary:

This study examines how individual movement vigor is integrated into a shared, dyadic vigor when two individuals are physically coupled. Participants performed wrist-reaching movements toward targets at different distances while mechanically linked via a virtual elastic band, and dyads were formed by pairing participants with different baseline vigor profiles. Under interaction conditions, movements converged to coordinated patterns that could not be explained by simple averaging, indicating that each dyad behaved as a single functional unit. Notably, under coupling, movement durations for both partners were shorter than in the solo condition, arguing against the view that each individual simply executed an independent movement plan. Furthermore, dyadic vigor was primarily predicted by the slower partner's vigor rather than by the faster partner's, suggesting that neither a leader-follower strategy nor a weighted averaging account fully explains the observed behavior. The authors propose a computational model in which both partners adapt to the emerging interaction dynamics ("interactive adaptation strategy"), providing a coherent explanation of the behavioral observations.

Strengths:

The study is carefully designed and addresses an important question about how individual movement vigor is integrated during joint action. The experimental paradigm allows systematic manipulation of interaction strength and partner asymmetry. The behavioral results show clear and robust patterns, particularly the shortening of movement durations under elastic coupling (KL and KH condition) and the asymmetrical contribution of the slower partner's vigor to dyadic vigor. The computational model captures the main behavioral patterns well and provides a principled framework for interpreting dyadic vigor not as a simple combination of two independent motor plans, but as an emergent property arising from mutual adaptation. Conceptually, the study is notable in extending the notion of vigor from an individual attribute to a dyad-level construct, opening a new perspective on coordinated movement and motor decision-making.

Weaknesses:

The revised manuscript now clearly explains why the proposed computational model successfully accounts for the observed dyadic behavior. In particular, the mechanisms by which uncertainty associated with the slower partner and time-related costs of the faster partner jointly shape dyadic vigor are now clear. I have no further comments to add.

<https://doi.org/10.7554/eLife.109781.2.sa2>

Reviewer #3 (Public review):

Summary:

This study provides novel insights into how individuals regulate the speed of their movements both alone and in pairs, highlighting consistent differences in movement vigor across people and showing that these differences can adapt in dyadic contexts. The findings are significant because they reveal stable individual patterns of action that are flexible when interacting with others, and they suggest that multiple factors, beyond reward sensitivity, may contribute to these idiosyncrasies. The evidence is generally strong, supported by careful behavioral measurements and appropriate modeling.

The authors have addressed all of my previous comments. I appreciate the clarification of abbreviations, terminology, and key concepts, the expansion of the discussion, and the

adjustments to some of the statistical analyses in response to both my earlier comments and those of Reviewer 1.

<https://doi.org/10.7554/eLife.109781.2.sa1>

Author response:

The following is the authors' response to the original reviews.

We thank the reviewers for their constructive and precise comments, which have helped us improve the consistency and clarity of our manuscript. Below, we provide a point-by-point response to each comment. In summary, the main changes introduced in the revised version are as follows:

- (1) We replaced all the statistical analyses to their non-parametric equivalents to ensure compliance with test assumptions and consistency of the results;
- (2) We compare the participants' reaction times before and during connected practice, revealing a significant reduction in reaction times of both partners when connected;
- (3) We added, in the supplementary materials, a table reporting the vigor scores of each participant in each experimental condition, facilitating the assessment of individual and dyadic behaviors;
- (4) We have reviewed and refined the terminology throughout the manuscript and reduced the number of abbreviations to improve clarity.

Public Reviews:

Reviewer #1 (Public review):

Summary:

The authors present a novel investigation of the movement vigor of individuals completing a synchronous extension-flexion task. Participants were placed into groups of two (so-called "dyads") and asked to complete shared movements (connected via a virtual loaded spring) to targets placed at varying amplitudes. The authors attempted to quantify what, if any, adjustments in movement vigor individual participants made during the dyadic movements, given the combined or co-dependent nature of the task. This is a novel, timely question of interest within the broader field of human sensorimotor control.

Participants from each dyad were labeled as "slow" (low vigor) or "fast" (high vigor), and their respective contributions to the combined movement metrics were assessed. The authors presented four candidate models for dyad interactions: (a) independent motor plans (i.e., co-activity hypothesis), (b) individual-led motor plans (i.e., leader-follower hypothesis), (c) generalization to a weighted average motor plan (i.e., weighted adaptation hypothesis), and (d) an uncertainty-based model of dynamic partner-partner interaction (i.e., interactive adaptation hypothesis). The final model allowed for dynamic changes in individual motor plans (and therefore, movement vigor) based on partner-partner interactions and observations. After detailed observations of interaction torque and movement duration (or vigor), the authors concluded that the interactive adaptation model provided the best explanation of human-human interaction during self-paced dyadic movements.

Strengths:

The experimental setup (simultaneous wrist extension-flexion movements) has been thoroughly vetted. The task was designed particularly well, with adequate block pseudo-randomization to ensure general validity of the results. The analyses of torque interaction, movement kinematics, and vigor are sound, as are the statistical measures used to assess significance. The authors structured the work via a helpful comparison of several candidate models of human-human interaction dynamics, and how well said models explained variance in the vigor of solo and combined movements. The research question is timely and extends current neuroscientific understanding of sensorimotor control, particularly in social contexts.

We thank the reviewer for their in-depth analysis and constructive assessment of our manuscript.

Weaknesses:

(1) My chief concern about the study as it currently stands is the relatively low number of data points (n=10). The authors recruited 20 participants, but the primary conclusions are based on dyad-specific interactions (i.e., analyses of "fast" vs "slow" participants in each pair). Some of these analyses would benefit greatly, in terms of power, from the addition of more data points.

We understand and appreciate the reviewer's concern regarding the effective sample size at the dyad level (n=10). While our primary analyses focus on dyad-specific interactions, we note that the reported effects are consistent across multiple dynamic conditions and are associated with large effect sizes. To provide a conservative assessment the Cohen's *D* values reported correspond to the smallest effect size observed across the relevant statistical tests, thereby limiting the risk of false positives or overinterpretation. In addition, to ensure robustness given the sample size and distribution properties of the data, we have replaced all parametric tests with their non-parametric counterparts, as some analyses violated ANOVA assumptions. Friedman and Kruskal-Wallis tests are now used for paired and unpaired main effects respectively, and Wilcoxon and Mann-Whitney tests for paired and unpaired post-hoc comparisons respectively. Note that these changes did not alter the conclusions of the study.

(a) The distribution of delta-vigor (Fast group vs Slow group) is highly skewed (see Figures 3D, S6D), with over half of the dyads exhibiting delta-vigor less than 0.2 (i.e., less than 20% of unit vigor). Given the relatively low number of dyads, it would be helpful for the authors to provide explicit listings of VigorFast, VigorSlow, and VigorCombined for each of the 10 separate dyads or pairings.

We agree with this comment. However, we note that the distribution of vigor scores within a population is typically centered around 1, with large deviations observed only for the fastest and slowest participants [1]. As a result, the distribution of Δ -vigor is inherently skewed. Correcting for this skewness would (i) require pairing participants based on their vigor, which is logistically difficult, and (ii) lead to an atypical sampling of dyads, with an over representation of pairs exhibiting very large vigor differences. The distributions of vigor scores for the fast and slow groups before and after the interaction are reported in Supplementary Fig. S21. In addition, as suggested by the reviewer, we have now included Table S.1 in the supplementary materials, listing the values VigorFast, VigorSlow, and VigorCombined for each of the 10 dyads. This table provides a complete view of the evolution of participant's vigor throughout the experiment.

(b) The authors concluded that the interactive adaptation hypothesis provided the best summary of the combined movement dynamics in the study. If this is indeed the case, then the relative degree of difference in vigor between the fast and slow participants in a dyad should matter. How well did the interactive adaptation model explain variance in the dyads with relatively low delta-vigor (e.g., less than 0.2) vs relatively high delta-vigor?

We initially expected the magnitude of difference in individual vigor within a dyad to play a significant role. However, our analysis did not reveal any systematic effect of Δ -vigor on either the interaction force or the resulting dyadic vigor, as shown by the LMM analysis. Importantly, the interactive adaptation hypothesis does per se imply that the magnitude of vigor differences between the two partners should matter, only that their respective roles in selecting the adapted behavior is different. Although the model includes several free parameters, we did not attempt to fit it to individual dyads as would in principle be possible. Instead, we performed a sensitivity analysis to assess how variations in the difference in vigor between the partners influence model predictions. For this purpose, we simulated increasing values of μ and variations in the fast partner's cost of time. In addition, we demonstrated that uncertainty in the estimated behavior of the slow partner, which is a priori specific to each individual, has a substantial impact on the optimal movement duration of the dyad. Overall, this analysis shows that the model captures the full range of qualitative trends observed in the experimental data. When applied to predict the behavior of the average dyad, the resulting movement time prediction error remain small, as detailed in the Results section.

(2) The authors shared the results of one analysis of reaction time, showing that the reaction times of the slow partners and the fast partners did not differ during the initial passive block. Did the authors observe any changes in RT of either the slow or fast partner during the combined (primary task) blocks (KL, KH, etc.)? If the pairs of participants did indeed employ a form of interactive adaptation, then it is certainly plausible that this interaction would manifest in the initial movement planning phase (i.e., RT) in addition to the vigor and smoothness of the movements themselves.

We thank the reviewer for this interesting question, that prompted us to extend our analysis of reaction times to the connected conditions. This additional analysis revealed a significant main effect of the condition on the reaction time for both the fast and slow groups (in both cases: $W_2 > 0.39$, $p < 0.02$). Post-hoc comparisons showed a significant reduction in reaction time between the initial null-field block (NF1) and the KH condition for the slow group ($p = 0.03$, $D = 1.46$), and a similar trend for the fast group ($p = 0.06$, $D = 1.03$). However, the reaction times remained comparable between the two groups, with no significant difference between them. We have incorporated these observations in the Results section (p.4, l.100–109) and expanded the Discussion (p.11, l.341–348) to address their implications for interactive adaptation in human-human and human-robot physical interactions.

Reviewer #2 (Public review):

Summary:

This study examines how individual movement vigor is integrated into a shared, dyadic vigor when two individuals are physically coupled. Participants performed wrist-reaching movements toward targets at different distances while mechanically linked via a virtual elastic band, and dyads were formed by pairing participants with different baseline vigor profiles. Under interaction conditions, movements converged to coordinated patterns that could not be explained by simple averaging, indicating that each dyad behaved as a single functional unit. Notably, under coupling, movement durations for both partners were shorter than in the solo condition, arguing against the view that each individual simply executed an independent movement plan. Furthermore, dyadic vigor was

primarily predicted by the slower partner's vigor rather than by the faster partner's, suggesting that neither a leader-follower strategy nor a weighted averaging account fully explains the observed behavior. The authors propose a computational model in which both partners adapt to the emerging interaction dynamics ("interactive adaptation strategy"), providing a coherent explanation of the behavioral observations.

Strengths:

The study is carefully designed and addresses an important question about how individual movement vigor is integrated during joint action. The experimental paradigm allows systematic manipulation of interaction strength and partner asymmetry. The behavioral results show clear and robust patterns, particularly the shortening of movement durations under elastic coupling (KL and KH conditions) and the asymmetrical contribution of the slower partner's vigor to dyadic vigor. The computational model captures the main behavioral patterns well and provides a principled framework for interpreting dyadic vigor not as a simple combination of two independent motor plans, but as an emergent property arising from mutual adaptation. Conceptually, the study is notable in extending the notion of vigor from an individual attribute to a dyad-level construct, opening a new perspective on coordinated movement and motor decision-making.

We thank the reviewer for their thorough analysis of our manuscript and their constructive feedback.

Weaknesses:

(1) A key conceptual issue concerns the apparent asymmetry between partners in the computational framework. While dyadic vigor is empirically better predicted by the slower partner's vigor, the model formulation appears to emphasize the faster partner's time-related cost and interaction forces. Although the cost function includes an uncertainty-related component associated with the slower partner, it remains unclear from the current formulation and description how dyadic vigor is formally derived from the slower partner's control policy within the same modeling framework. This raises an important question regarding whether the model offers a symmetric account of dyadic vigor formation for both partners or whether it is effectively anchored to the faster partner's control architecture.

We have modified our phrasing to clarify the principles according to which the computational framework was designed (p.7, l.226–231 and p.9, l.260–264). As stated in the Results section, the model is indeed asymmetric by design, which corresponds to the different roles of the fast and slow partner exhibited in the data. In that context, the uncertain term associated with the slow partners should be understood as an overarching constraint that conditions the strategy of the dyad, while the fast partner cost of time acts as a contributor to the expected dyad strategy. Conceptually and numerically as reported in the sensitivity analysis, this asymmetry corresponds to the role of the slow partners in setting the vigor ranking among the dyads and the role of the fast partner in setting the average dyadic behavior.

(2) A second conceptual issue concerns the interpretation of the term "motor plan." It remains unclear whether this term refers primarily to movement-related characteristics such as speed or duration, or more broadly to the underlying optimization structure that governs these variables. This distinction is theoretically important, as it determines whether the reported interaction effects should be understood as adjustments in movement characteristics or as changes in the structure of the control policy itself.

We agree with the reviewer that this terminology required clarification. In this paper, the

term “motor plan” refers to the time series of control inputs planned by the CNS, rather than solely to kinematic descriptors such as speed or duration. These planned control signals are a direct consequence of the underlying optimization structure and cost functions that govern trajectory generation. We have clarified this definition in the Introduction (p.1, 1.23–24).

Reviewer #3 (Public review):

Strengths:

This study provides novel insights into how individuals regulate the speed of their movements both alone and in pairs, highlighting consistent differences in movement vigor across people and showing that these differences can adapt in dyadic contexts. The findings are significant because they reveal stable individual patterns of action that are flexible when interacting with others, and they suggest that multiple factors, beyond reward sensitivity, may contribute to these idiosyncrasies. The evidence is generally strong, supported by careful behavioral measurements and appropriate modeling, though clarifying some statistical choices and including additional measures of accuracy and smoothness would further strengthen the support for the conclusions.

Thank you for this analysis and the insightful feedback.

Major Comments:

(1) Given the idiosyncrasies in individual vigor, would linear mixed models (LMMs) be more appropriate than ANOVAs in some analyses (e.g., in the section "Solo session"), as they can account for random intercepts and slopes on vigor measures? Some figures (e.g., Figure 2.B and 3.E) indeed seem to show that some aspects of behaviour may present variability in slopes and intercepts across participants. In fact, I now realize that LMMs are used in the "Emergence of dyadic vigor from the partners' individual vigor" section, so could the authors clarify why different statistical approaches were applied depending on the sections?

We thank the reviewer for this thoughtful comment. We deliberately used different statistical approaches throughout the paper in order to address different types of questions. Note that the statistical tests were converted to their nonparametric equivalent for consistency (see answer to Reviewer 1).

- Friedman tests were used in a limited number of cases to assess population- or group-level effects, such as differences in movement time, smoothness, or accuracy across the solo, connected, and after-effects conditions. Such tests provide a straightforward framework for these descriptive, condition-level comparisons.

- The stability of individual and dyadic vigor scores across conditions was assessed using Pearson correlations across all condition pairs, which we consider the most direct and interpretable approach for evaluating consistency across sessions.

- LMMs were employed to examine how dyadic vigor relates to the partners' individual vigor measured in the solo conditions, which revealed the critical contribution of the slow partner.

Rather than applying a single statistical framework throughout, we selected the method best suited to each question. While LMMs are well suited for modeling participant-specific variability when linking individual and dyadic measures, their systematic use in all analyses would be less intuitive and would not directly address several of the population-level comparisons central to this study.

(2) If I understand correctly, the introduction suggests that idiosyncrasies in movement vigor may be driven by interindividual differences in reward sensitivity. However, the current task does not involve any explicit rewards, yet the authors still observe

idiosyncrasies in vigor, which is interesting. Could this indicate that other factors contribute to these consistent individual differences? For example, could sensitivity to temporal costs or physical effort explain the slow versus fast subgrouping? Specifically, might individuals more sensitive to temporal costs move faster to minimize opportunity costs, and might those less sensitive to effort costs also move faster? Along the same lines, could the two subgroups (slow vs. fast) be characterized in terms of underlying computational "phenotypes," such as their sensitivities to time and effort? If this is not feasible with the current dataset, it would still be valuable to discuss whether these factors could plausibly account for the observed patterns, based on existing literature.

We thank the reviewer for this interesting question. We first note that the notion of reward in motor control is quite broad. Although our task did not include explicit external (e.g. monetary) rewards, we assumed that participants attribute an implicit value to completing the task in accordance with the experimenter's instructions. This assumption has been shown to be appropriate for characterising baseline behavior in previous studies [2–5].

As discussed in the Introduction, vigor is generally understood to emerge from a tradeoff between effort, accuracy, and time. The reviewer is correct in noting that inter-individual differences in vigor may reflect differences in reward sensitivity or in its discounting [3,6], given that time and reward are intrinsically coupled. Differences in vigor may also arise from inter-individual variability in sensitivity to effort or perceived task difficulty. Because these factors are intertwined—for example, increasing accuracy through co-contraction typically incurs greater effort [7])—it is challenging to disentangle their respective contributions based solely on behavioral data.

In the present study, our inverse optimal control procedure to identify the cost of time (and thus predict individuals' vigor) relies on a predefined effort-accuracy tradeoff under fixed final time across multiple movement amplitudes [8]. As a result, the model does not allow us to independently estimate individual sensitivities to effort, accuracy, and time. Such characterization of computational "phenotypes" would likely require experimental paradigms in which each of these factors is systematically manipulated while the others are held constant, which is beyond the scope of the current dataset. In practice, the main value of behavioral modeling lies in revealing the relative weighting of these criteria by the CNS during motor planning [5]. We have expanded the Discussion to clarify these limitations and considerations (see Discussion p.12, l.396–401 & l.407–412).

Finally, we chose not to emphasize these broader issues in the present manuscript because (i) they are peripheral to our primary research question on how individual vigor influences human-human interaction, and (ii) although we do not yet have definitive and consensual answers, they have been addressed in multiple studies reviewed elsewhere [9,10].

(3) The observation that dyads did not lose accuracy or smoothness despite changes in vigor is interesting and suggests a shift in the speed-accuracy tradeoff. Could the authors include accuracy and smoothness measures in the main figures rather than only in supplementary materials? I think it would make the manuscript more complete.

We also find that the preservation of accuracy and smoothness despite changes in vigor is an interesting result, and we therefore chose to report these measures in the Supplementary Materials. However, we believe it is preferable not to include them in the main figures for the following reasons:

- We avoid framing our results in terms of a speed-accuracy trade-off, as Fitts' work was initially designed to study fast movements [11], whereas our work focuses on self-paced movements. As outlined in the Introduction, vigor is more appropriately interpreted as reflecting a tradeoff between effort (related to movement speed), accuracy, and time. From

this perspective, the reported changes of vigor already capture a shift in the underlying trade-off selected by the CNS, using a framework better suited to our experimental paradigm.

- The manuscript is technically dense and reports multiple analyses that are essential to establish (i) the existence and definition of dyadic vigor, and (ii) how it emerges from interaction between partners. Although the observed preservation of accuracy and improvements in smoothness are informative, they are not central to these two primary questions and would risk diverting attention from the core contributions of the paper. In addition, accuracy is not a feature predicted by our deterministic modeling and extensions would be needed to capture these aspect. Here we only attempted to replicate average behaviors.

(4) It is a bit unclear to me whether the variance assumptions for ANOVAs were checked, for instance, in Figure 3H.

We thank the reviewer for this comment, which prompted us to verify the assumptions underlying our ANOVAs. We found that a few distributions in the original analysis, as well as in some of the new tests, did not meet these assumptions. To ensure consistency, all statistical analyses have now been replaced with non-parametric tests: Friedman and Kruskal-Wallis tests for paired and unpaired main effects, Wilcoxon and Mann-Whitney tests for paired and unpaired post-hocs. The updated results do not change any of the conclusions. the only minor change is accuracy, that appeared slightly improved in a restricted number of connected conditions, and now appears mostly non-impacted.

Recommendations for the authors:

Reviewer #1 (Recommendations for the authors):

Minor points:

(1) Lines 146-147. The authors state, "Whereas the fast partners maintained a similar duration". Figures S6H,I suggest that fast partners made slower movements during the paired task relative to the solo task, not movements with a similar duration.

We agree that Fig. S.6H,I suggest slightly slower movements for the fast partners, though not significant. We have modified the sentence to be less assertive than in the previous version (see p.6, l.155).

(2) In the Discussion (Lines 318-319), the authors state that their findings confirm and extend the "benefits of dyadic control in collaborative actions". What benefits are they referring to here, relative to individual control? It would be helpful if the authors would elaborate on this claim.

We have modified this sentence to clarify that the benefits of dyadic control refer to previously reported advantages over individual control, namely reduced movement time Reed and Peshkin (2008) [12] and improved tracking accuracy [13,14] (see p.11, l.336–337).

(3) On Lines 87-89, the authors reference a decomposition of variance of vigor scores across the NF1, VL, and VH conditions; however, I did not see an explanation of how this decomposition was performed. The method used to estimate variance explained by inter-individual vs intra-individual differences in vigor should be outlined for the reader.

Thank you for pointing out this missing information. We now explain in the statistical analysis section (see p.14, l.504–507), that the percentage of inter-individual variability in vigor is estimated using sum-square values as an estimation of inter- and intra-individual variability.

$$\text{Var}_{\text{inter}} = 100 \frac{\text{SS}_{\text{inter}}}{\text{SS}_{\text{inter}} + \text{SS}_{\text{intra}}} \quad (1)$$

(4) How was the absolute interaction torque for a paired movement calculated? Was it an integral of the temporal profile of torque for some portion of the combined movement? The method for calculating the absolute interaction torque needs to be specified.

We have now clarified in the Methods (see p.14, l.490–491) that the reported average interaction effort was computed as the absolute value of the interaction torque as a function of time averaged over the entire movement.

(5) Lines 123–124: "... interaction torque showed no significant correlation with differences in individual vigor within dyads." This statement should be supported by appropriate statistical measures.

This result is now supported by reporting the corresponding Pearson correlation analyses. No significant correlations were found between interaction torque and differences in individual vigor within dyads (KL conditions: $|r| < 0.43$, $p > 0.22$; KH conditions: $|r| < 0.18$, $p > 0.61$, see p.5, l.132–133).

(6) For the analysis, presented in Figure 3C, and specified on lines 116–123, the text mentions the main effects of both condition and target. There doesn't appear to be much of an effect of the target for the KH data. Should these results not be reported as an interaction effect between the two factors instead?

We agree with the reviewer and have corrected our presentation of these results (see p.4, l.126–128). Consistent with the reviewer's observation, no significant effect of the target is found in the KH condition.

(7) Figures 3E and S6B. What is the purpose of including the averaged data for each pair in addition to both individuals' data from each pair? It would be useful to distinguish the individual data from the average data for each pair. Frankly, the number of data points shown on this sub-figure is excessive.

There may have been a misunderstanding. Because the partners of a dyad are connected by a virtual elastic band (rather than a rigid bar), they do not execute identical movements. Therefore Figs. 3E, S6B display the movement time of all individual participants, together with the corresponding 20 individual regression lines, like in Fig. 2B. The solid black line represents the average across all individuals, and the averaged behaviors of dyads are not included. We have clarified this point by revising the caption of Fig. 3E (see p.5).

Noted mis-spellings:

Figure S.3A caption: "trials towards this target."

Page 10 Line 313: "Importantly, these findings show ...".

These mis-spellings have been corrected at supplementary p.2 and main text p.11, l.331. Thank you!

Reviewer #2 (Recommendations for the authors):

(1) To illustrate the contribution of the three components used to calibrate the overall cost function, it would be informative to include simulation analyses in which each component is selectively removed (i.e., ablation analyses).

We did not perform ablation analyses, as selectively removing components of the model can lead to instability or ill-suited control inputs, making the resulting simulations difficult to

interpret. Instead, we conducted a sensitivity analysis of the key parameters shaping the overall cost function, including the estimated mean and deviation of the slow partner's movement duration, the weight associated with uncertain torque minimization (Figs. S.18,S.19), and the fast partner's cost of time (Fig. S20). This analysis reveals the predominant roles of the estimated slow partner movement patterns in determining the model predictions, in agreement with our experimental observations.

(2) Although the authors refer to the motor-off condition as "passive," participants actively generated the movements in the absence of external forces. Thus, this condition corresponds to active, unassisted movement. A different term may therefore reduce potential confusion for readers.

We agree that term "passive" was not well-chosen given the context of the paper, thus we have instead replaced this denomination as "null-field" condition. Consequently, the P1 and P2 blocks are now referred to as NF1 and NF2.

(3) Please clarify the instructions given to participants. Were they informed in advance that their movements would physically interact with those of their partner?

Thank you for pointing out this missing clarification. We have now specified in the Methods (p.14, l.465–469) that participants were not informed prior to any condition that they would interact with a human partner; they were only told that the robot would provide assistance. When debriefed at the end of the experiment, only one out of the 20 participants reported having realized that they were connected to another human. Most participants believed they were interacting either with a version of themselves or with a robot with some randomness.

(4) Line 475. Should "Fig. 2D" be "Fig. 2B"?

Thank you for catching this error. The reference has been corrected to Fig. 2B (see p.15, l.522).

Reviewer #3 (Recommendations for the authors):

(1) The analysis of reaction times shows no difference between groups in the passive block, which challenges the assumption that movement vigor covaries with decision speed or action initiation speed. It may be worth discussing this in the context of recent literature.

We agree that the initial analysis and discussion of reaction times were too superficial. In the revised manuscript, we now report that dyadic interaction leads to significantly shorter reaction times (p.4, l.100–109), concomitantly with improved movement velocity. We have also expanded the Discussion, on the relationship between decision and action speeds/durations (p.11, l.340–348).

(2) Many abbreviations are unusual for a non-expert. I would recommend using the full terms instead. At least initially, I found it difficult to follow the results because the abbreviations were not immediately clear (at least to me).

We agree that the paper had too many abbreviations. Therefore, we have removed the abbreviated names of the models and, when possible without impacting the readability, used the full names of the conditions.

(3) Relatedly, the notation in Figure 1 may be confusing. The labels "S" and "F" (slow and fast) correspond to different concepts than "F" and "L" (follower and leader), so the same participant could be labeled "F" as fast but not "F" as a leader.

Thank you for pointing out this potential source of confusion. We have therefore modified Fig. 1A (p.2) to avoid any potential confusion by using the full model names rather than abbreviations. In the remainder of the manuscript, "S" and "F" exclusively denote the slower

and faster partners within a dyad, and we do not use abbreviations for "leader" or "follower" in the text.

(4) In figures like 2.C and 3.I, keeping the same scales on the x and y axes and adding a diagonal reference line would make it easier to see shifts across conditions.

As explained in the Methods, vigor scores in the low- and high-viscosity conditions were computed using the average movement durations from the NF1 condition as a reference. Consequently, because movements are slower in these conditions, the corresponding vigor values are lower than those in NF1. For this reason, using identical scales on the x- and y-axes and adding a 45° reference line could mislead the reader in thinking that the vigor scores are expected to be identical and reduce the readability of the figure.

(5) Multiple hypotheses about dyadic regulation of vigor are nicely explained; it could help to indicate if any of these were a priori favored based on prior literature.

Previous literature provides mixed evidence regarding how vigor might be regulated in dyadic interaction. For instance, Takagi et al. (2016) [15] reported that mechanically connected partners may rely on independent motor plans, which corresponds to the co-activity hypothesis considered here. However, in that study, movement duration was prescribed. We therefore expected that removing this constraint on movement duration could allow coordination strategies to emerge, particularly in view of findings on haptic communication during tracking of random targets while connected via an elastic band [13,14].

At the same time, a large body of work on human–human and human–robot interaction has interpreted coordination through a leader–follower framework. In our context, vigor is understood as the outcome of a tradeoff between effort and elapsed time, with time being associated with a decaying reward. Based on this framework, we hypothesized a priori that a leader–follower scheme would emerge, in which the fast partner—being more sensitive to time costs and/or less sensitive to effort—would tend to drive the interaction, even at the expense of increased effort. For these reasons, the leader–follower hypothesis was formulated as the expected outcome throughout the manuscript.

(6) In the introduction, statements such as "relative vigor of an individual is remarkably stable" appear true only in the solo condition. The same is true in the discussion where it is said that vigor is a stable trait. The whole study show that an individual can shift his/her vigor to the same vigor of another individual, so it doesn't appear stable to me in such conditions but adaptable.

Let us first clarify that when we describe vigor as "remarkably stable", we do not imply that individuals do not adjust their movement timing in response to changes in external dynamics. For example, movement durations increase in visco-resistive conditions even during solo performance; nevertheless, individuals who move faster in the absence of resistance will remain faster relative to others when resistance is introduced. In this sense, stability refers to the preservation of relative rankings across conditions, rather than invariance of absolute movement timing. Because interaction with another individual constitutes a substantial change in task dynamics, an effect on individual pace is therefore expected.

Told that (and as pointed to by the reviewer) (i) dyadic interactions lead to the emergence of a dyadic vigor characterized by average movement durations close to those of the fast partners, while the ranking across dyads is largely imposed by the slow partners; and (ii) these adaptations persist after the interaction phase. Importantly, the observed vigor adaptations appear to last longer in our physical interaction task than in previous attempts to manipulate vigor using visual feedback [16]. To account for this adaptability of vigor, we have (i) clarified

claims in the Introduction regarding the stability of vigor (see p.1, l.18–20), and (ii) expanded the Discussion to more explicitly address vigor adaptability and the possible resulting consequences for the concept of vigor (see p.12, l.407–412).

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<https://doi.org/10.7554/eLife.109781.2.sa0>